Chapter 3
Quality Control

3.1. Statistical Quality Control refers to the application of descriptive statistics and inferential statistics to the production process in order to reduce defects or waste, while at the same time maintaining high-quality and uniform products.

● **<u>The statistical quality control procedure</u>** can be summarized as follows:

1. Randomly select a sample of n units during each of the k production intervals within a single production process.
2. Use the data obtained in Step 1 to construct control charts, such as the $\bar{X}$ - chart, R-chart, S-chart, P-chart or C-chart

The choice of control chart depends on the nature of the data, whether the data are quantitative or qualitative.

3. Interpret the control chart to determine whether the production process at that stage is in control.

● **<u>The data recording format for the k production intervals</u>**

Time period	The values representing the characteristics of the n sampled units				Statistical value
	1	2	…	n	
1	X_{11}	X_{12}	…	X_{1n}	w_1
2	X_{21}	X_{22}	…	X_{2n}	w_2
⋮			⋮		⋮
k	X_{k1}	X_{k2}	…	X_{kn}	w_k

X_{ij} denote the characteristic of the product sampled in time period i for sample unit j ; where $i = 1, 2, \ldots, k$ and $j = 1, 2, \ldots, n$

<u>Objectives of Using a Control Chart</u>

1. To examine whether each stage of the production process shows any abnormality or change.
2. To identify the causes of abnormalities in the production process and determine corrective actions to ensure proper operation.

● **<u>Components of a Control Chart</u>**

1. <u>Central Line (CL)</u>

   The Central Line represents the mean of the sample statistics calculated from the k samples.

   $\bar{\bar{X}}$ denotes the mean of the sample means.
   $\bar{R}$ denotes the mean of the sample ranges.

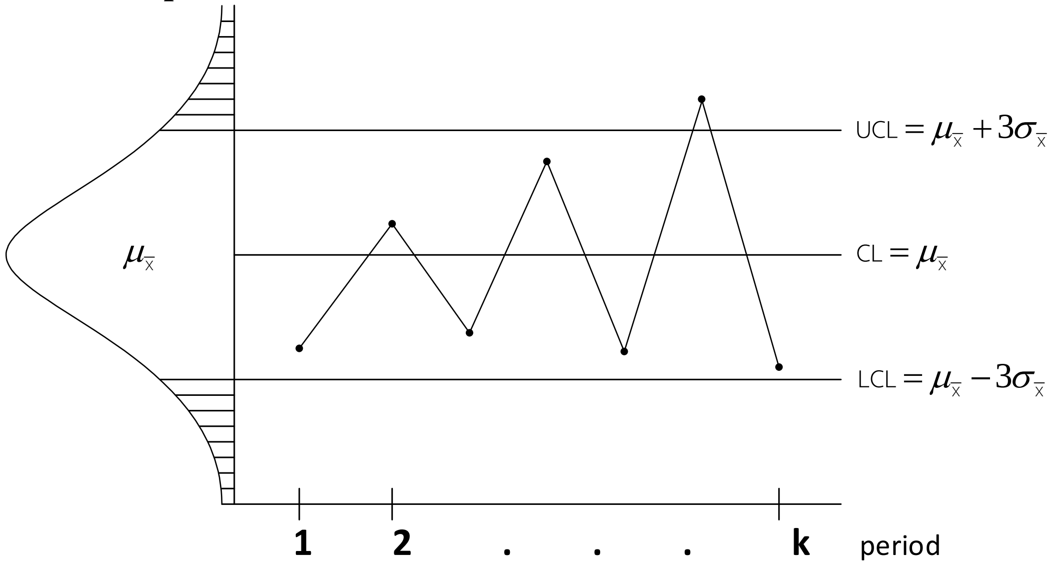
   $\bar{S}$ denotes the mean of the sample standard deviations.
   $\bar{P}$ denotes the mean of the sample proportions of defects.

   $\bar{C}$ denotes the mean number of defects per unit of product in the sample.

2. <u>Upper Control Limit (UCL) and Lower Control Limit (LCL)</u>

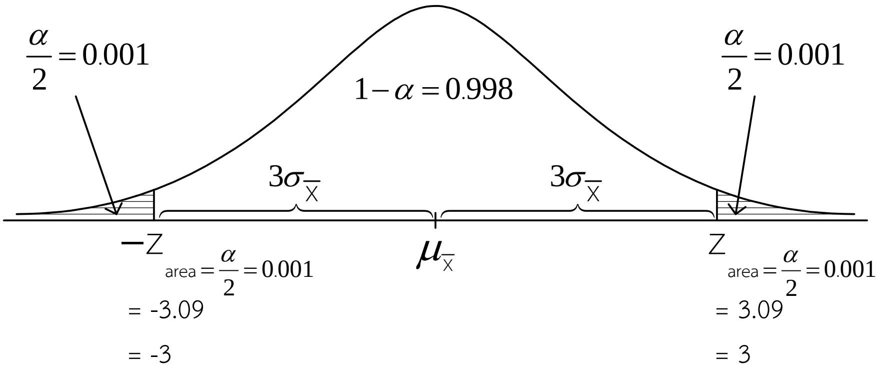
   These are straight lines positioned at a distance of three times the standard deviation of the statistic being monitored from the Central Line (CL), whether it is $\bar{\bar{X}} \pm 3\sigma_{\bar{X}}$, $\bar{P} \pm 3\sigma_P$, $\bar{R} \pm 3\sigma_R$, $\bar{C} \pm 3\sigma_C$, $\bar{S} \pm 3\sigma_S$

   Let $\bar{X}_1, \bar{X}_2, \ldots, \bar{X}_k$ the sample mean computed from the k samples across the k production intervals.



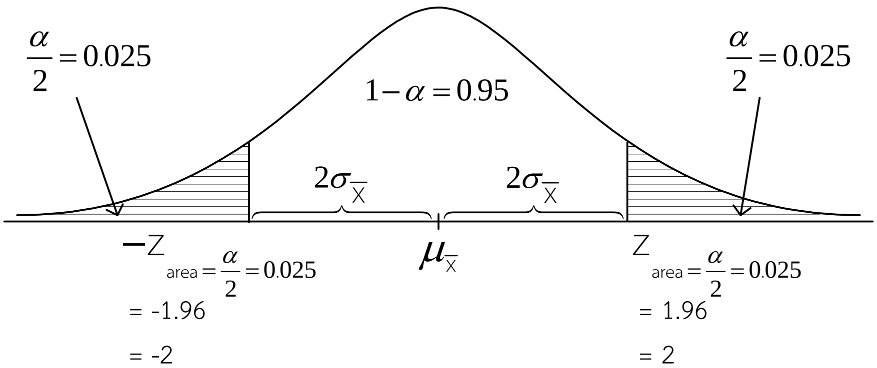
- **<u>Out of Control</u>** refers to a production process that is not in control (i.e., at least one point lies outside the UCL or LCL).
- **<u>In Control</u>** refers to a production process that is in control (i.e., all points lie within the UCL and LCL, and the pattern of the points shows no trend and appears random). In this case, the constructed chart can be used as the control chart for monitoring product quality.
- **<u>There are three types of Control Chart formats:</u>**

1) <u>Three-Sigma Control Chart</u> (3σ) refers to an interval estimate at the 99.8%
confidence level.



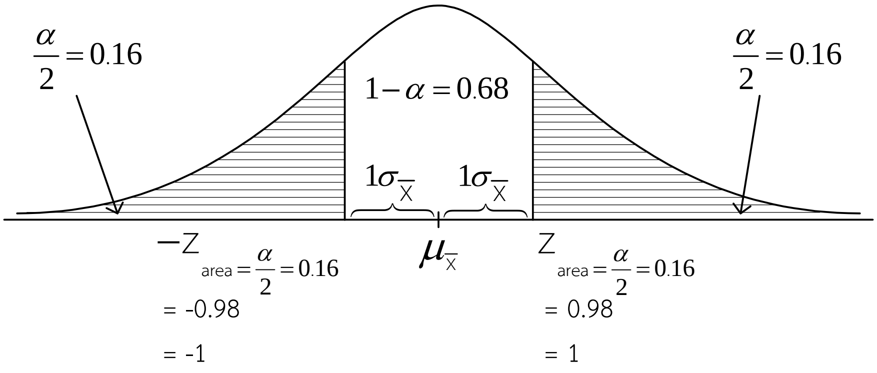
Therefore, the 99.8% confidence interval for $\overline{X}$ (also called the 3σ interval) is $\mu_{\overline{X}} \pm 3\sigma_{\overline{X}}$

2) <u>Two-Sigma Control Chart</u> (2σ) refers to an interval estimate at the 95% confidence level.



Therefore, the 95% confidence interval for $\overline{X}$ (also called the 2σ interval) is $\mu_{\overline{X}} \pm 2\sigma_{\overline{X}}$

3) <u>One-Sigma Control Chart</u> (1σ) refers to an interval estimate at the 68% confidence level.



Therefore, the 68% confidence interval for $\bar{X}$ (also called 1σ interval) is $\mu_{\bar{X}} \pm 1\sigma_{\bar{X}}$

- **<u>Causes of Variation</u>** In a production process, various types of variation may occur. These variations can be classified into two categories:

  1) <u>Chance Variation</u>

This refers to variation that occurs randomly and cannot be completely eliminated from the production process. Examples include:

- humidity in the air
- the amount of dust in the air
- weather conditions
- background noise, and so on.

2) <u>Assignable Variation</u>

This refers to variation that arises from identifiable and controllable causes. It is non-random and can be reduced or eliminated from the production process. Examples include:

- defects or malfunctioning of machines
- changes in the raw materials used in production
- inconsistency or variation in the raw materials
- the skill level of the machine operator, and so on.

Therefore, each stage of the production process must be inspected using statistical quality control techniques to identify the causes and defects that occur, determine where they arise, and take corrective actions to eliminate or minimize such variation.

3.2 Types of Control Charts

In constructing charts to monitor product quality, the choice of which type of control chart to use depends on the nature of the data collected from the production process, specifically, whether the data are quantitative or qualitative.

Control charts can be classified into two types:

1. Control Charts for Quantitative Data These are used when the data collected can be measured numerically, such as weight, length, width, thickness, durability, and similar characteristics.

2. Control Charts for Qualitative Data These are used when the data collected are based on product attributes or characteristics that reflect defects, such as nonconforming items or damaged products.

3.2.1 Control Charts for Variables

These charts are used when the quality of a product can be measured numerically; for example, weight, length, volume, or hardness. The commonly used control charts in this category are the $\bar{X}$ -Chart, R-Chart and S-Chart.

A sample of size n is drawn from the production process during each sampling interval, with the intervals spaced evenly throughout the k sampling periods.

sampling interval (number of subgroups)	Sample size						
	1	2	…	n	$\bar{X}_i$	R_i	S_i
1	X_{11}	X_{12}	…	X_{1n}	$\bar{X}_1$	R_1	S_1
2	X_{21}	X_{22}	…	X_{2n}	$\bar{X}_2$	R_2	S_2
⋮			⋮			⋮	
k	X_{k1}	X_{k2}	…	X_{kn}	$\bar{X}_k$	R_k	S_k

X_{ij} denotes the characteristic of the product sampled in time interval i for sample unit j ; where $i=1,2,\dots,k$ and $j=1,2,\dots,n$

$\bar{X}_i=\frac{\sum_{j=1}^{n}X_{ij}}{n}$ = is the sample mean of subgroup i

$R_i = \max_j(X_{ij}) - \min_j(X_{ij})$ = is the sample range of subgroup i

S_i = The sample standard deviation of subgroup i can be computed using two formulae, as follows:

1) $S_i = \sqrt{\dfrac{\sum_{j=1}^{n}(X_{ij}-\bar{X}_i)^2}{n-1}}$ 2) $S_i = \sqrt{\dfrac{1}{n-1}\left(\sum_{j=1}^{n}X_{ij}^2 - \dfrac{\left(\sum_{j=1}^{n}X_{ij}\right)^2}{n}\right)}$

Let $\bar{X}_1, \bar{X}_2, \ldots, \bar{X}_k$ be the sample mean in time interval $1,2,\ldots,k$ respectively,

and $\bar{\bar{X}}$ = the mean of the sample means, which can be computed using two formulae:

1) $\bar{\bar{X}} = \dfrac{\sum_{i=1}^{k}\bar{X}_i}{k} = \dfrac{\bar{X}_1+\bar{X}_2+\ldots+\bar{X}_k}{k}$

2) $\bar{\bar{X}} = \dfrac{\sum_{i=1}^{k}\sum_{j=1}^{n}X_{ij}}{nk} = \dfrac{X_{11}+X_{12}+\ldots+X_{k1}+\ldots+X_{kn}}{nk}$

Note There are two formulae for the computation. The choice of which formula to use depends on the requirements specified in the problem.

Let $R_1, R_2, \ldots, R_k$ be the sample range in time interval $1,2,\ldots,k$ respectively,

and $\bar{R}$ = the mean of the sample ranges.

$$\bar{R} = \frac{\sum_{i=1}^{k}R_i}{k} = \frac{R_1+R_2+\ldots+R_k}{k}$$

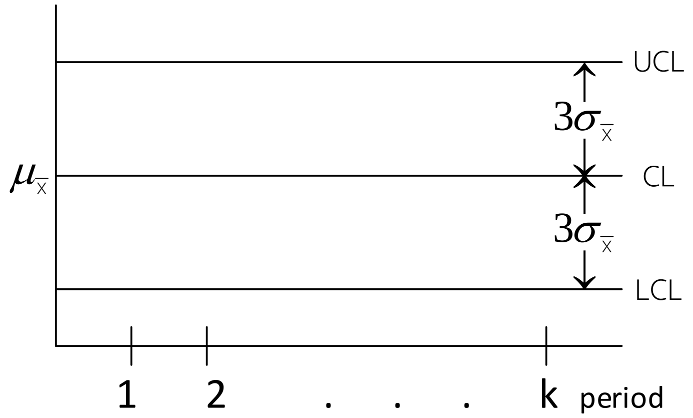
Let $S_1, S_2, \ldots, S_k$ be the sample standard deviation in time interval $1,2,\ldots,k$ respectively, and $\bar{S}$ = the mean of the sample standard deviations.

$$\bar{S} = \frac{\sum_{i=1}^{k} S_i}{k} = \frac{S_1 + S_2 + \ldots + S_k}{k}$$

- **$\bar{X}$ - Chart (Mean Control Chart)**

refers to a chart constructed to display the variation that may occur in a production process, using $\bar{X}$ as the standard quality indicator when the process is in control.

The 3-sigma mean control chart for $\bar{X}$ is $\mu_{\bar{X}} \pm 3\sigma_{\bar{X}}$



In practice, the value of $\mu_{\bar{X}}$ is unknown, so it is estimated by $\bar{\bar{X}}$ This yields the $3\sigma_{\bar{X}}$ interval of $\bar{X}$ that is

$$\text{UCL} = \bar{\bar{X}} + 3\sigma_{\bar{X}}$$
$$\text{CL} = \bar{\bar{X}}$$
$$\text{LCL} = \bar{\bar{X}} - 3\sigma_{\bar{X}}$$

In practice, the value of $\sigma_{\bar{X}}$ is unknown, and there are 2 methods for estimating it as follows:

1) Estimate $\sigma_{\bar{X}}$ using $\frac{\bar{R}}{d_2}\left(\frac{1}{\sqrt{n}}\right)$ This method is called "$\bar{X}$-Chart Based on R" this gives

$$\text{UCL} = \bar{\bar{X}} + 3\frac{\bar{R}}{d_2}\left(\frac{1}{\sqrt{n}}\right) = \bar{\bar{X}} + 3\frac{\bar{R}}{d_2}\left(\frac{1}{\sqrt{n}}\right)$$
$$\text{CL} = \bar{\bar{X}}$$
$$\text{LCL} = \bar{\bar{X}} - 3\frac{\bar{R}}{d_2}\left(\frac{1}{\sqrt{n}}\right) = \bar{\bar{X}} - 3\frac{\bar{R}}{d_2}\left(\frac{1}{\sqrt{n}}\right)$$

where $A_2 = \frac{3}{d_2}\left(\frac{1}{\sqrt{n}}\right)$

A_2 is obtained from the Control Chart Constant table, such as

$n=5$, $A_2 = 0.577$

$n=20$, $A_2 = 0.180$

2) Estimate $\sigma_{\bar{X}}$ using $\frac{\bar{S}}{c_4}\left(\frac{1}{\sqrt{n}}\right)$ .This method is called “ $\bar{X}$ -Chart Based on S” This gives

$$\boxed{\begin{aligned} \text{UCL} &= \bar{\bar{X}} + 3\frac{\bar{S}}{c_4}\left(\frac{1}{\sqrt{n}}\right) = \bar{\bar{X}} + A_3\bar{S} \\ \text{CL} &= \bar{\bar{X}} \\ \text{LCL} &= \bar{\bar{X}} - 3\frac{\bar{S}}{c_4}\left(\frac{1}{\sqrt{n}}\right) = \bar{\bar{X}} - A_3\bar{S} \end{aligned}}$$

Where $A_3 = \frac{3}{c_4}\left(\frac{1}{\sqrt{n}}\right)$

A_3 is obtained from the Control Chart Constant table, such as

$n=5$, $A_3 = 1.427$

$n=20$, $A_3 = 0.680$

Example 3.1 In the production of a certain type of product, the factory produces for 7 hours per day, and an equal number of product samples, 10 pieces, are randomly selected in each production hour. Specify the values of n and k and find the values of A_2 and A_3

Solution In this case, the sample size of the products randomly selected in each hour (n) = 10

and the number of time intervals sampled every hour consecutively for 7 hours (k) = 7

Therefore, from the Control Chart Constant table ; $n=10$, $n=10$, $A_2=0.308$

Example 3.2 Product samples are randomly selected for 9 hours. The sample size in each hour is equal, and the total number of samples obtained is 63 pieces. Specify the values of n and n and find the values of k and n

Solution

In this case, the sample size of the products randomly selected in each hour (n) = total samples / k = 63 / 9 = 7

and the number of time intervals sampled every hour consecutively for 9 hours (k) = 9

Therefore, from the Control Chart Constant table ; n = 7 , n = 7

A_2 = 0.419 , A_3 = 1.182 , D_3 = 0.076 , D_4 = 1.924

B_3 = 0.118 , B_4 = 1.882 , d_2 = 2.704

Example 3.3 A factory has installed a new machine used for filling 41-ounce canned coffee, but variation is expected to occur. The quality control department randomly selected 5 cans every hour for 4 consecutive hours and weighed them, in ounces. The data obtained are as follows

Time interval	Can number					$\bar{X}_i$	R_i	S_i
	1	2	3	4	5			
8.00	41	43	42	41	43	$\bar{X}_1 = 42$	$R_1 = 2$	$S_1 = 1.00000$
9.00	39	40	40	39	42	$\bar{X}_2 = 40$	$R_2 = 3$	$S_2 = 1.22474$
10.00	41	44	43	46	41	$\bar{X}_3 = 43$	$R_3 = 5$	$S_3 = 2.12132$
11.00	38	39	40	39	39	$\bar{X}_4 = 39$	$R_4 = 2$	$S_4 = 0.70711$
						$\sum_{i=1}^{4} \bar{X}_i = 164$	$\sum_{i=1}^{4} R_i = 12$	$\sum_{i=1}^{4} S_i = 5.05317$

Construct the control chart for the mean ($\bar{X}$ -Chart)

Solution Let X_{ij} denote the weight (ounces, or "oz.") of the canned coffee in hour i , can number j

$$; \quad i = 1, 2, \ldots, 4 \qquad , \qquad j = 1, 2, \ldots, 5$$

The computation of the sample mean of the coffee weight in each hour is as follows

$$\bar{X}_1 = \frac{\sum_{j=1}^{n} X_{1j}}{n} = \frac{41+43+42+41+43}{5} = \frac{210}{5} = 42 \text{ oz.}$$

$$\bar{X}_2 = \frac{\sum_{j=1}^{n} X_{2j}}{n} = \frac{39+40+40+39+42}{5} = \frac{200}{5} = 40 \text{ oz.}$$

$$\bar{X}_3 = \frac{\sum_{j=1}^{n} X_{3j}}{n} = \frac{41+44+43+46+41}{5} = \frac{215}{5} = 43 \text{ oz.}$$

$$\bar{X}_4 = \frac{\sum_{j=1}^{n} X_{4j}}{n} = \frac{38+39+40+39+39}{5} = \frac{195}{5} = 39 \text{ oz.}$$

The computation of the sample standard deviation of the coffee weight in each hour is as follows

$$S_1 = \sqrt{\frac{\sum_{j=1}^{n}(X_{1j}-\bar{X}_1)^2}{n-1}}$$
$$= \sqrt{\frac{(41-42)^2+(43-42)^2+(42-42)^2+(41-42)^2+(43-42)^2}{5-1}} = 1 \text{ oz.}$$

Alternatively, compute S_1 from the dispersion formula as follows

$$S_1 = \sqrt{\frac{1}{n-1}\left(\sum X_1^2 - \frac{\left(\sum X_1\right)^2}{n}\right)} = \sqrt{\frac{1}{5-1}\left(8,824-\frac{(210)^2}{5}\right)} = 1 \text{ oz.}$$

Therefore, the sample standard deviation of the coffee weight in each hour is as follows

$S_1 = 1.00000$ oz.

$S_2 = 1.22474$ oz.

$S_3 = 2.12132$ oz.

$S_4 = 0.70711$ oz.

And the sample range of the coffee weight in each hour is as follows

$R_1 = 43-41 = 2$ oz.

$R_2 = 42-39 = 3$ oz.

$R_3 = 46-41 = 5$ oz.

$R_4 = 40-38 = 2$ oz.

Using the sample means of the coffee weight in each hour, compute **the mean of the sample means** as follows

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k}\bar{X}_i}{k} = \frac{\sum_{i=1}^{4}\bar{X}_i}{4} = \frac{\bar{X}_1+\bar{X}_2+\bar{X}_3+\bar{X}_4}{4} = \frac{42+40+43+39}{4} = \frac{164}{4} = 41 \text{ oz.}$$

or compute the mean of the sample means using the following formula

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k}\sum_{j=1}^{n} X_{ij}}{nk} = \frac{\sum_{i=1}^{4}\sum_{j=1}^{5} X_{ij}}{nk} = \frac{X_{11} + X_{12} + \ldots + X_{41} + \ldots + X_{45}}{5 \times 4}$$

$$= \frac{41 + 43 + \ldots + 38 + \ldots + 39}{5 \times 4} = \frac{820}{5 \times 4} = 41 \text{ oz.}$$

Using the sample ranges of the coffee weight in each hour, compute **the mean of the sample ranges** as follows

$$\bar{R} = \frac{\sum_{i=1}^{k} R_i}{k} = \frac{\sum_{i=1}^{4} R_i}{4} = \frac{R_1 + R_2 + R_3 + R_4}{4} = \frac{2+3+5+2}{4} = \frac{12}{4} = 3 \text{ oz.}$$

Using the sample standard deviations of the coffee weight in each hour, compute **the mean of the sample standard deviations** as follows.

$$\bar{S} = \frac{\sum_{i=1}^{k} S_i}{k} = \frac{\sum_{i=1}^{4} S_i}{4} = \frac{S_1 + S_2 + S_3 + S_4}{4} = \frac{1 + 1.22474 + 2.12132 + 0.70711}{4}$$

$$= \frac{5.05317}{4} = 1.2633 \text{ oz.}$$

Constructing the control chart for the mean using the $\bar{X}$ - Chart (Based on R) has formulae for calculating the Upper Control Limit (UCL) and Lower Control Limit (LCL) as follows

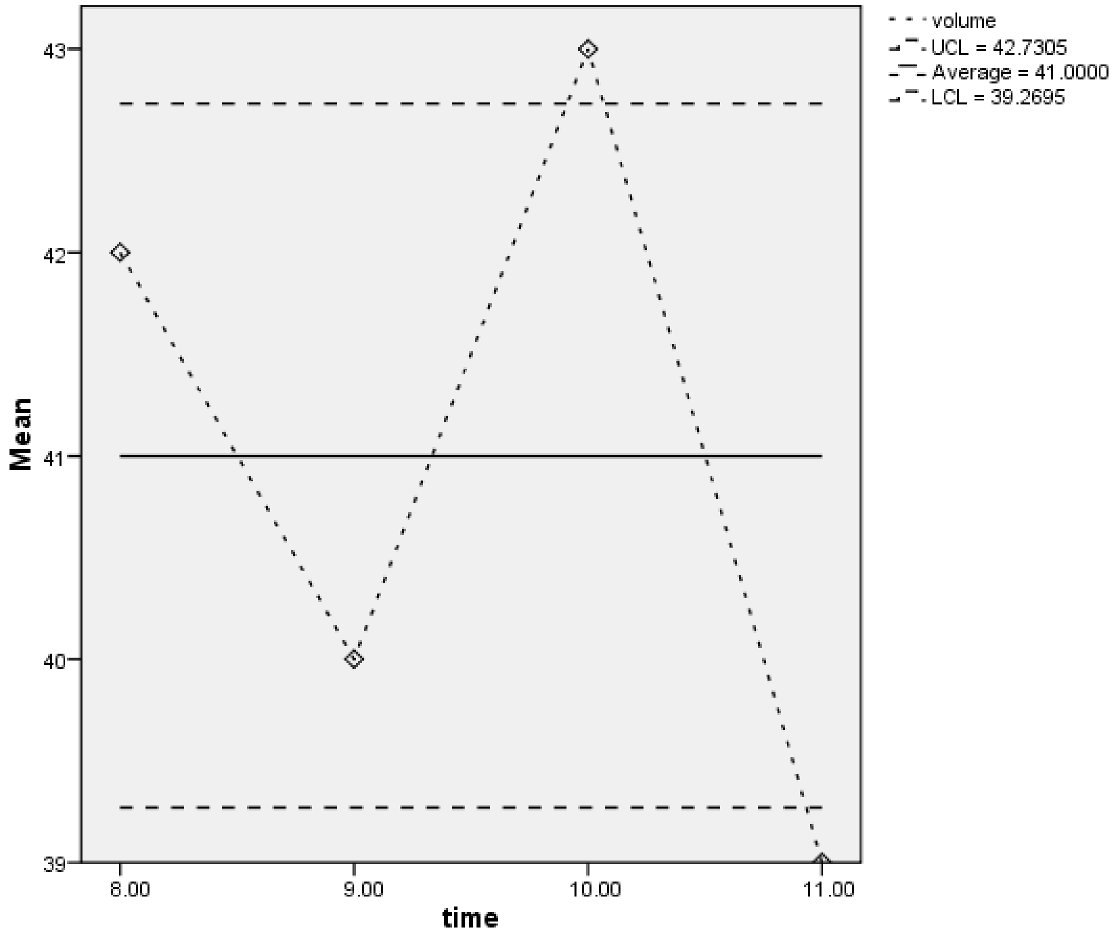
$\bar{X}$ - Chart (Based on R)

UCL = $\bar{\bar{X}} + A_2\bar{R}$ = $41 + 0.577(3)$ = 42.73 ออนซ์

CL = $\bar{\bar{X}}$ = 41 oz.

LCL = $\bar{\bar{X}} - A_2\bar{R}$ = $41 - 0.577(3)$ = 39.27 ออนซ์

In this case $n = 5$, $A_2 = 0.577$



Constructing the control chart for the mean using the $\bar{X}$ - Chart (Based on S) has formulae for calculating the Upper Control Limit (UCL) and Lower Control Limit (LCL) as follows

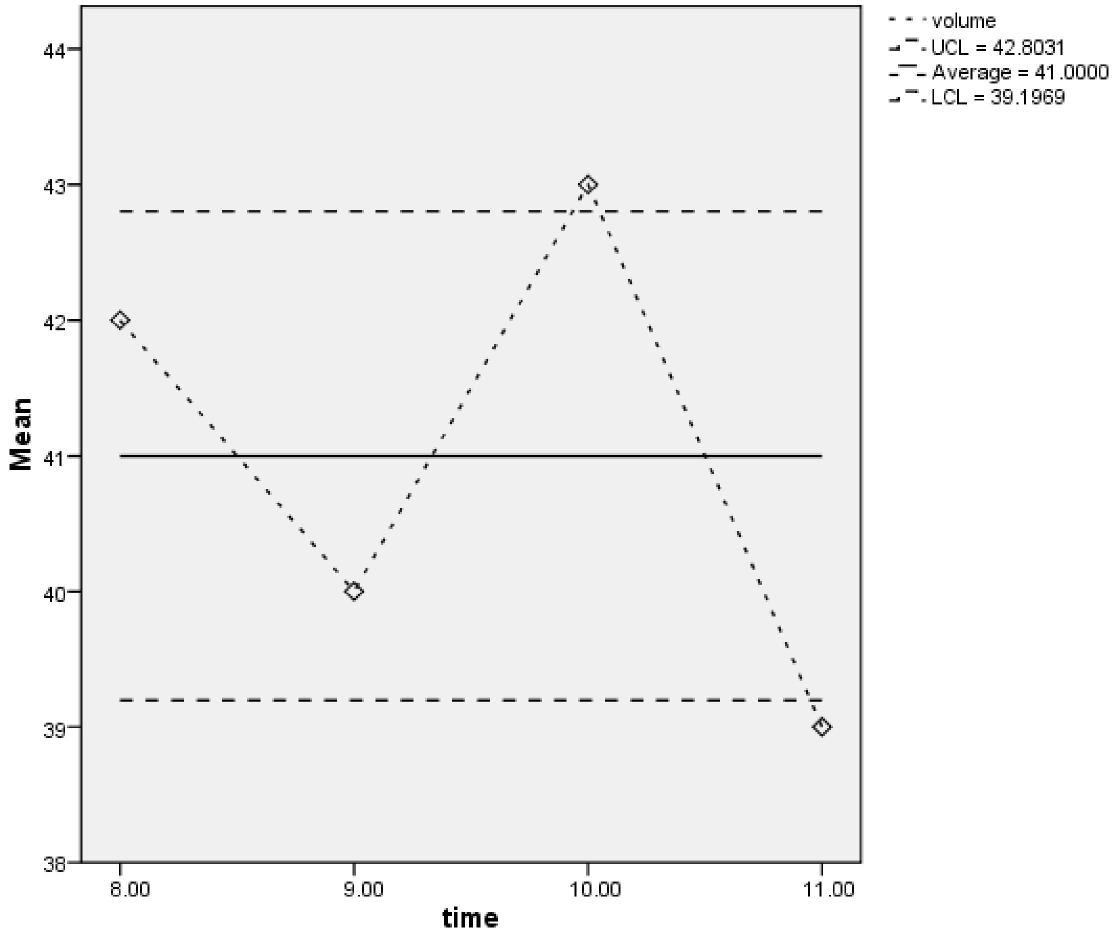
$\bar{X}$ - Chart (Based on S)

UCL = $\bar{\bar{X}} + A_3\bar{S}$ = $41+1.427(1.2633)$ = 42.80 oz.

CL = $\bar{\bar{X}}$ = 41 oz.

LCL = $\bar{\bar{X}} - A_3\bar{S}$ = $41-1.427(1.2633)$ = 39.20 oz.

In this case $n=5$, $A_3 = 1.427$

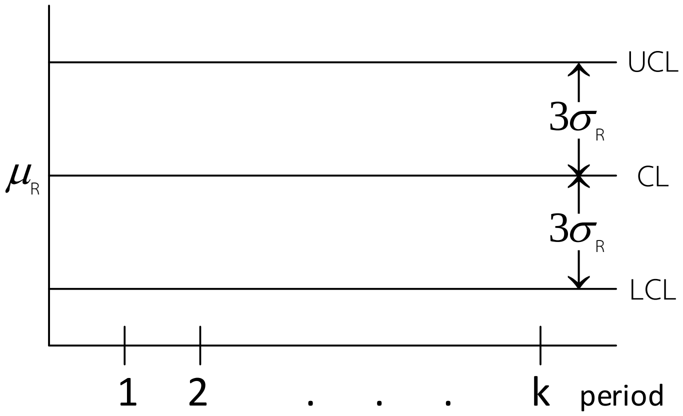


From the figure, the $\bar{X}$ **- Chart (Based on R)** and $\bar{X}$ **- Chart (Based on S)** shows that the sample means at 10:00 a.m. and 11:00 a.m. are outside the UCL and LCL, respectively. Therefore, it can be concluded that the average weight of the canned coffee is **Out of Control**.

- **<u>Range Control Chart</u> (Range Chart : R - Chart)**

means a chart constructed to display the variation occurring in the ranges of different sample sets randomly selected for inspection when the production process is in control.

<u>The 3-Sigma Range Control Chart of R</u> is $\mu_R \pm 3\sigma_R$



In practice, the value of μ_R **is unknown**, so it is estimated by $\bar{R}$. This yields the $3\sigma_R$ interval of R as

UCL = $\bar{R}+3\sigma_R$
CL = $\bar{R}$

LCL = $\bar{R}-3\sigma_R$

In practice, the value of σ_R **is unknown**; estimate σ_R using σ_R.This gives

$$\text{UCL} = \bar{R}+3\frac{d_3}{d_2}\bar{R} = \bar{R}+3\frac{d_3}{d_2}\bar{R} = D_4\bar{R}$$
$$\text{CL} = \bar{R}$$
$$\text{LCL} = \bar{R}-3\frac{d_3}{d_2}\bar{R} = \bar{R}-3\frac{d_3}{d_2}\bar{R} = D_3\bar{R}$$

Where $D_3 = 1-3\frac{d_3}{d_2}$, $D_4 = 1+3\frac{d_3}{d_2}$

D_3 and D_3 obtained from the Control Chart Constant table

for example $n=5$, $D_3=0$, $D_4=2.115$

$n=20$, $D_3=0.414$, $D_4=1.586$

<u>Note:</u> If R – Chart has LCL < 0, use LCL = 0 instead.

Example 3.4 From Example 3.3, construct **the Range Control Chart (R – Chart)**

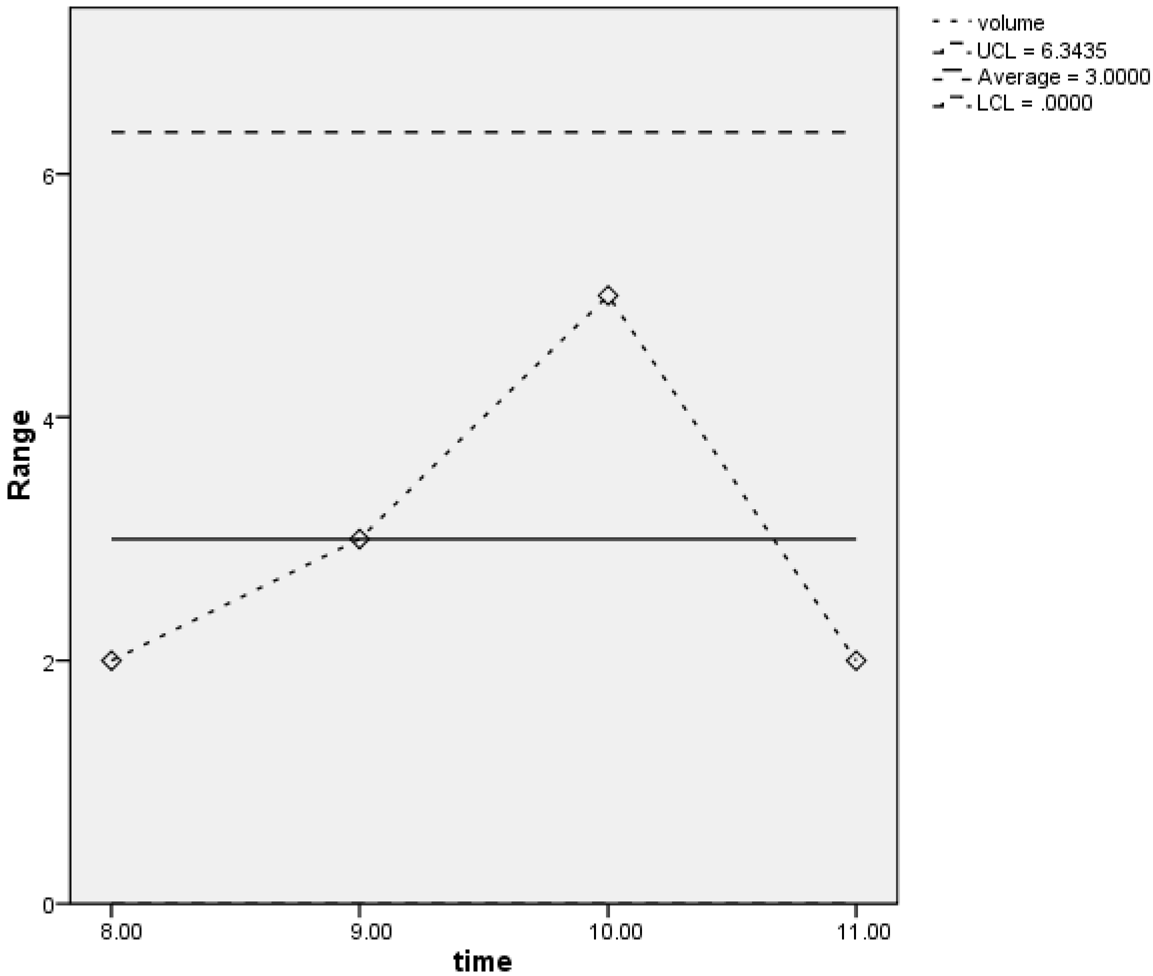
Solution Constructing the Range Control Chart (R – Chart) uses formulae for calculating the Upper Control Limit (UCL) and Lower Control Limit (LCL) as follows

UCL = $D_4\bar{R}$ = $2.115(3)$ = 6.34 oz.

CL = $\bar{R}$ = 3 oz.

LCL = $D_3\bar{R}$ = $0(3)$ = 0 oz.

In this case $n = 5$, $D_3 = 0$, $D_4 = 2.115$

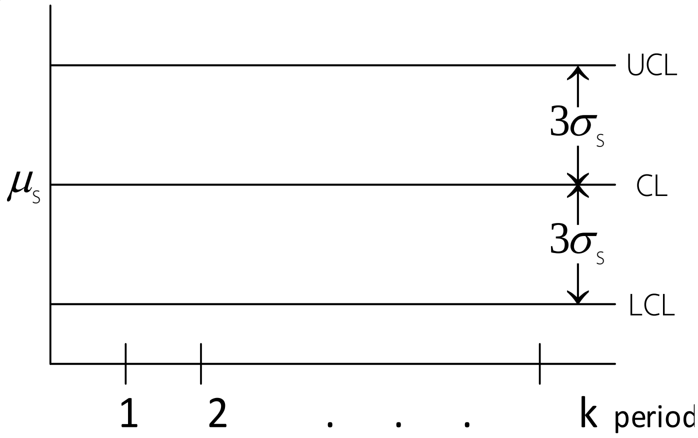


From the R - Chart, it is found that none of the sample ranges in any hour lies outside the UCL and LCL. In other words, the sample ranges of the canned coffee weights in each hour are dispersed within the boundaries of the UCL and LCL, indicating that the dispersion of the weights of the canned coffee sampled at various times is **in control (In Control).**

- **<u>Standard Deviation Control Chart</u> (Standard Deviation Chart: S - Chart)**

This means a chart constructed to examine the dispersion of the production process by using the standard deviation (S) as the indicator of quality when the production process is in control.

<u>The 3-Sigma Standard Deviation Control Chart of S</u> is $\mu_S \pm 3\sigma_S$



In practice, the value of μ_S **is unknown**, so it is estimated by $\bar{S}$. This yields the $3\sigma_S$ interval of S is

UCL = $\bar{S} + 3\sigma_S$

CL = $\bar{S}$

LCL = $\bar{S} - 3\sigma_S$

In practice, the value of σ_S **is unknown;** estimate σ_S using $\left[\sqrt{1-c_4^2}\right]\frac{\bar{S}}{c_4}$. This gives

UCL = $\bar{S}+3\left[\sqrt{1-c_4^2}\right]\dfrac{\bar{S}}{c_4}$ = $\dfrac{1}{c_4}\left[c_4+3\sqrt{1-c_4^2}\right]\bar{S}$ = $B_4\bar{S}$

CL = $\bar{S}$

LCL = $\bar{S}-3\left[\sqrt{1-c_4^2}\right]\dfrac{\bar{S}}{c_4}$ = $\dfrac{1}{c_4}\left[c_4-3\sqrt{1-c_4^2}\right]\bar{S}$ = $B_3\bar{S}$

Where $B_3 = \dfrac{1}{c_4}\left[c_4-3\sqrt{1-c_4^2}\right]$, $B_4 = \dfrac{1}{c_4}\left[c_4+3\sqrt{1-c_4^2}\right]$

B_3 and B_4 obtained from the Control Chart Constant table

for example $n=5$, $B_3=0$, $B_4=2.089$

$n=20$, $B_3=0.510$, $B_4=1.490$

Note 1) Since S **cannot be negative** if LCL < 0 use LCL = 0 instead.

2) R – Chart and S – Chart are usually constructed using only one of them, because they produce the same conclusion.

Example 3.5 From Example 3.3, construct **the Standard Deviation Control Chart (S – Chart)**

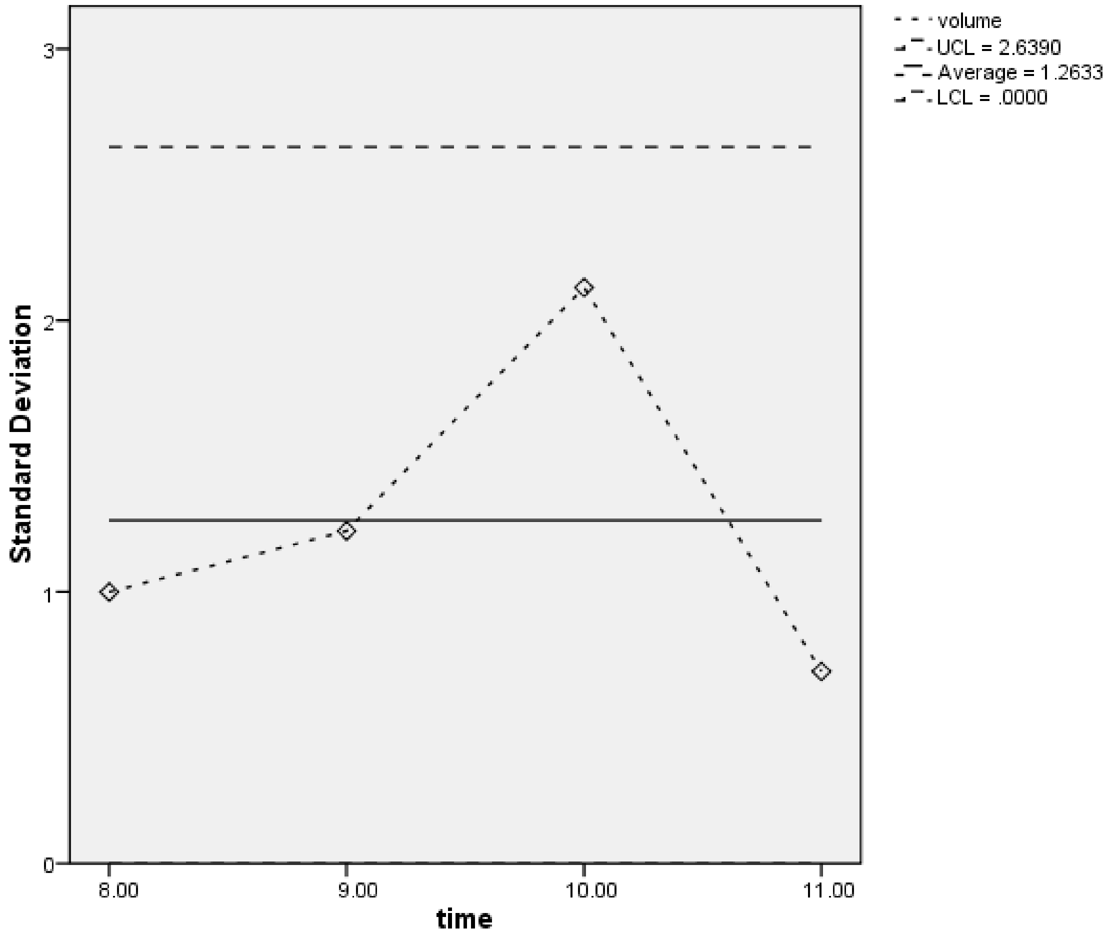
Solution Constructing the Standard Deviation Control Chart (S – Chart) uses formulae for calculating the Upper Control Limit (UCL) and the Lower Control Limit (LCL) as follows

UCL = $B_4\bar{S}$ = $2.089(1.2633)$ = 2.64 oz.

CL = $\bar{S}$ = 1.2633 oz.

LCL = $B_3\bar{S}$ = $0(1.2633)$ = 0 oz.

In this case $n=5$, $B_3=0$, $B_4=2.089$



From the S – Chart, it is found that none of the sample standard deviations in any hour lies outside the UCL and LCL. In other words, the sample standard deviations of the canned coffee weights in each hour are dispersed within the boundaries of the UCL and LCL, indicating that the dispersion of the weights of the canned coffee sampled at various times is **in control (In Control).**

- **To determine whether the production process is in control for quantitative data** it is divided into two approaches as follows:
 Approach 1: Consider the $\bar{X}$ – Chart (Based on R) together with the R – Chart or
 Approach 2: Consider the $\bar{X}$ – Chart (Based on S) together with the S–Chart

If considering **Approach 1**, and either one of the charts, namely the $\bar{X}$ – Chart (Based on R) or the R–Chart is not in control, it indicates that the entire production process is **not in control.**

Or, for **Approach 2**, if either one of the charts, namely the $\bar{X}$ – Chart (Based on S) or the S–Chart is not in control, it indicates that the entire production process is **not in control.**

- **Out of Control** when considering the $\bar{X}$ – Chart may be caused by
    1. machine installation
    2. operating temperature
    3. atmospheric humidity
    4. machine operator
- **Out of Control** when considering the R–Chart (or S–Chart) may be caused by
    1. inconsistency of materials used in production
    2. malfunctioning machinery
    3. lack of production knowledge or fatigue of the worker

Therefore, those involved must investigate the causes and take corrective action.

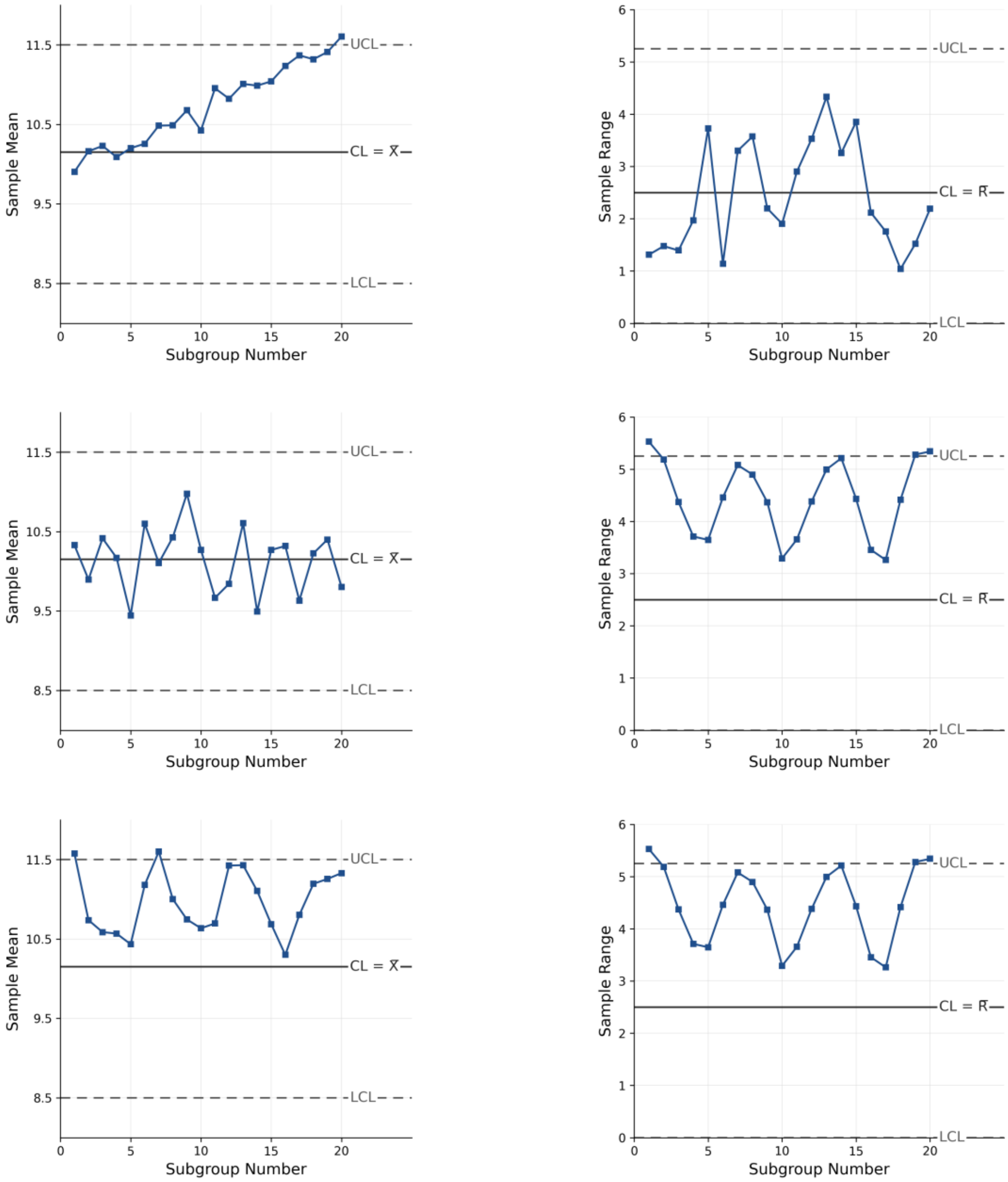
Note 1. If the constructed control chart shows a random distribution of points, that is, all points are scattered around the Central Line (CL) without any pattern and without any increasing trend, it indicates that the production process is in control (In Control).

2. If the constructed control chart shows a **non–random** distribution of points, in which some points may be scattered far away from the CL and/or some points may lie outside the UCL and LCL, before deciding that the production process is out of control (Out of Control), the analyst may remove those outlying points and construct a new control chart, and then make the decision as to whether the production process in that step is in control or out of control.

- **<u>Patterns that cause the production process to be out of control</u>**

1. Materials used in production are not consistent

<u>$\bar{X}$ – Chart</u> <u>R – Chart</u>



Example 3.6 A packaged rice distribution company received customer complaints regarding the packing weight not matching the stated amount of 5 kilograms per bag. The company's quality control manager therefore constructed control charts to inspect the production process

in the automatic rice–packing stage. Sample rice bags were randomly selected, 15 bags every 1 hour, and the data obtained are as follows

Sample set number	Average weight of rice	Range
1	4.90	0.4
2	4.95	0.5
3	5.00	0.4
4	4.90	0.6
5	4.95	0.4
6	4.60	0.6
7	4.90	0.5
8	5.10	0.4
9	4.80	0.6
10	4.90	0.5

Construct the control charts and decide whether the production process is in control.

Solution In this case, the sample size of the packaged rice bags randomly selected in each hour is (n) = 15 bags.

The number of time intervals sampled every hour consecutively for 10 hours is (k) = 10 intervals.

Let X_{ij} denote the weight of a packaged rice bag (in kilograms) in hour i and bag number j

; $i = 1, 2, 3, \ldots, 10$ Hour ; $j = 1, 2, \ldots, 15$ bag

From the given data, the sample means of the weights of the packaged rice bags in each hour are as follows.

$\bar{X}_1 = 4.90$ kilogram ; $\bar{X}_6 = 4.60$ kilogram

$\bar{X}_2 = 4.95$ kilogram ; $\bar{X}_7 = 4.90$ kilogram

$\bar{X}_3 = 5.00$ kilogram ; $\bar{X}_8 = 5.10$ kilogram

$\bar{X}_4 = 4.90$ kilogram ; $\bar{X}_9 = 4.80$ kilogram

$\bar{X}_5 = 4.95$ kilogram ; $\bar{X}_{10} = 4.90$ kilogram

From the given data, the sample ranges of the weights of the packaged rice bags in each hour are as follows.

$R_1 = 0.4$ kilogram	;	$R_6 = 0.6$ kilogram	
$R_2 = 0.5$ kilogram	;	$R_7 = 0.5$ kilogram	
$R_3 = 0.4$ kilogram	;	$R_8 = 0.4$ kilogram	
$R_4 = 0.6$ kilogram	;	$R_9 = 0.6$ kilogram	
$R_5 = 0.4$ kilogram	;	$R_{10} = 0.5$ kilogram	

Using the sample means of the weights of the packaged rice bags in each hour, compute **the mean of the sample means** as follows.

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{k}\bar{X}_i}{k} = \frac{\sum_{i=1}^{10}\bar{X}_i}{10} = \frac{\bar{X}_1 + \bar{X}_2 + \ldots + \bar{X}_{10}}{10} = \frac{49}{10} = 4.90 \text{ kilogram}$$

Using the sample ranges of the weights of the packaged rice bags in each hour, compute **the mean of the sample ranges** as follows.

$$\bar{R} = \frac{\sum_{i=1}^{k}R_i}{k} = \frac{\sum_{i=1}^{10}R_i}{10} = \frac{R_1 + R_2 + \ldots + R_{10}}{10} = \frac{4.9}{10} = 0.49 \text{ kilogram}$$

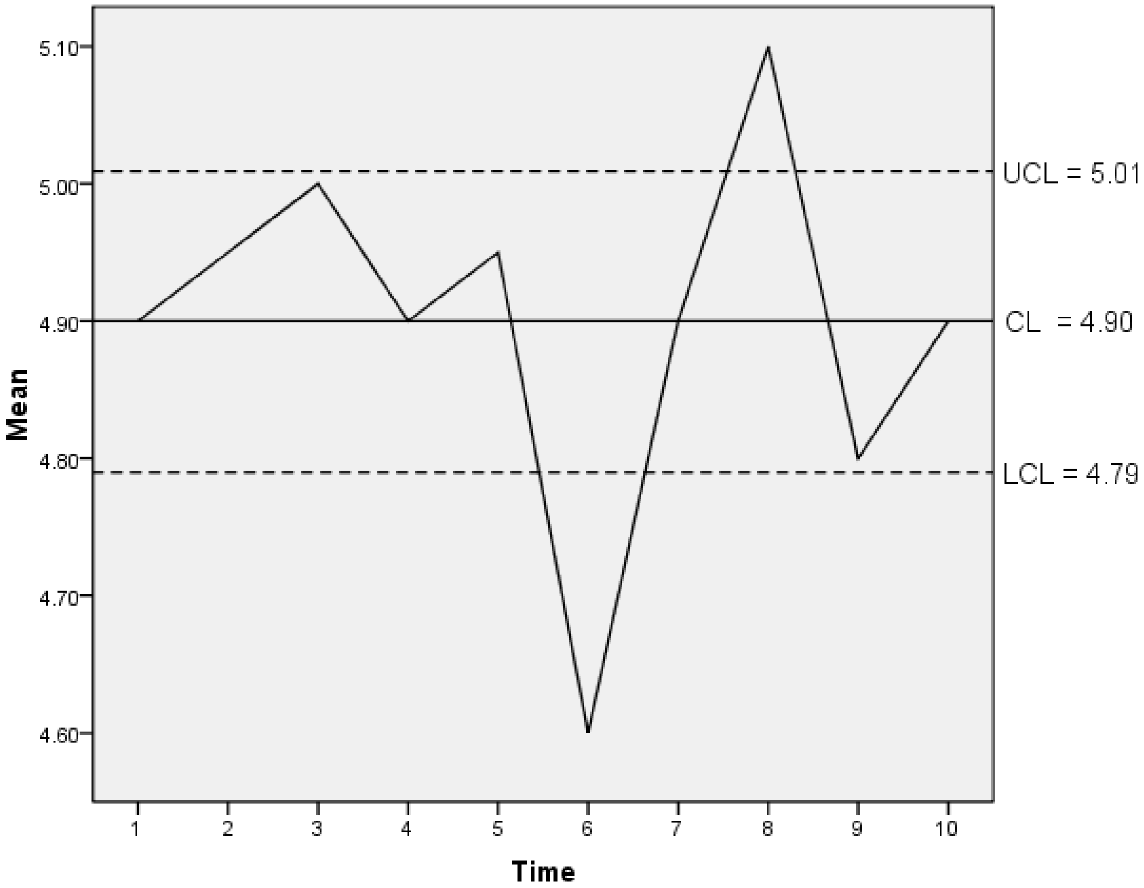
1. **<u>$\bar{X}$ - Chart (Based on R)</u>**

UCL = $\bar{\bar{X}} + A_2\bar{R}$ = $4.90 + 0.223(0.49)$ = 5.01 kilogram

CL = $\bar{\bar{X}}$ = 4.90 kilogram

LCL = $\bar{\bar{X}} - A_2\bar{R}$ = $4.90 - 0.223(0.49)$ = 4.79 kilogram

In this case $n = 15$, $A_2 = 0.223$



From the control chart for the mean, it is found that in production hours 6 and 8, the rice packing process is **out of control.** This may be caused by some type of variation. The production may be stopped for inspection, or the data in production hours 6 and 8 may be discarded and a new control chart may be constructed.

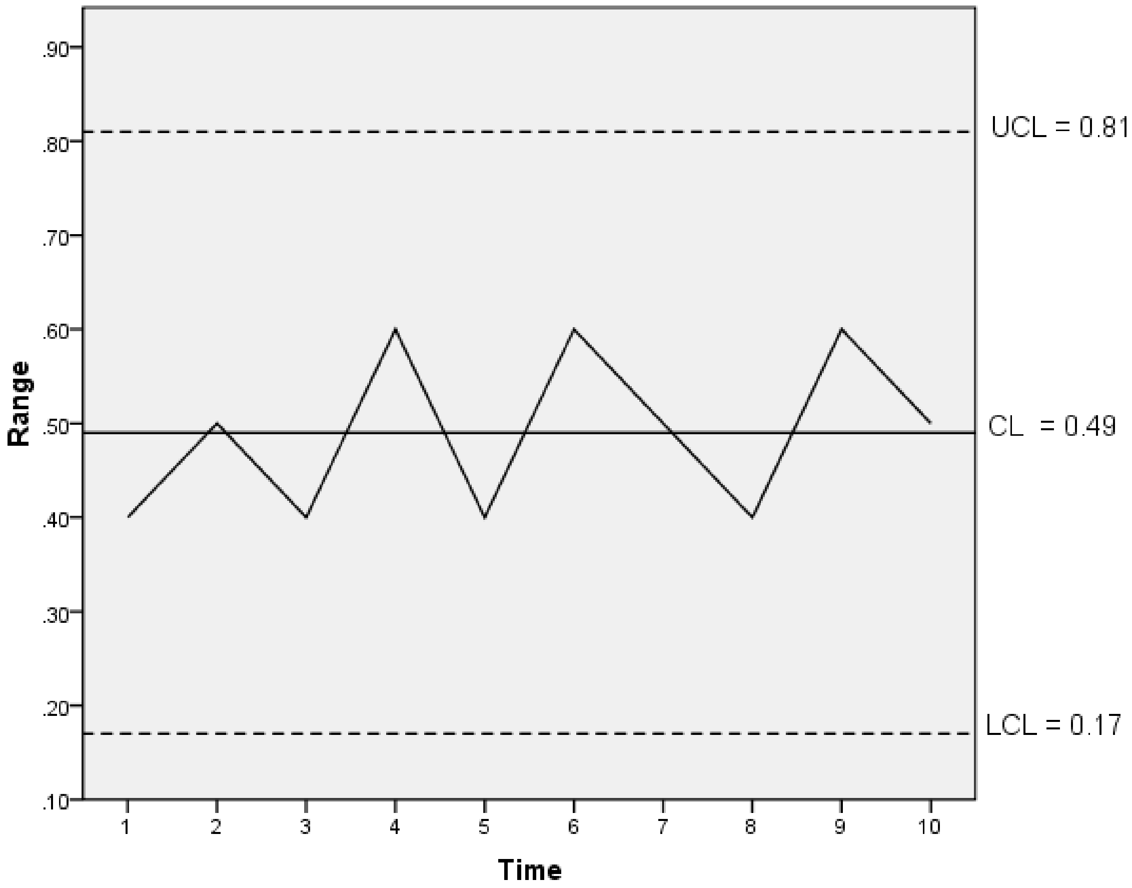
2. **R - Chart** (Consider the ranges of the 10 production hours)

UCL = $D_4\bar{R}$ = $1.652(0.49)$ = 0.81 oz.

CL = $\bar{R}$ = 0.49 oz.

LCL = $D_3\bar{R}$ = $0.348(0.49)$ = 0.17 oz.

In this case $n=15$, $D_3 = 0.348$, $D_4 = 1.652$



From the range control chart (considering the ranges of the 10 production hours), it is found that the distribution of the ranges of the weights of the packaged rice bags is **In Control.**

Conclusion

From Figure 1. $\bar{X}$ - Chart (Based on R). **(Out of Control)**

and Figure 2. R – Chart **(In Control)**

It can be concluded that the rice packing process using an automatic machine is **out of control.**

I. **$\bar{X}$ - Chart (Based on R)**: Calculated by discarding the data in production hours 6 and 8.

$$\bar{\bar{X}}_{NEW} = \frac{\sum_{i=1}^{8} \bar{X}_i}{8} = \frac{49-4.6-5.1}{8} = \frac{39.3}{8} = 4.91 \text{ kilogram}$$

$$\bar{R}_{NEW} = \frac{\sum_{i=1}^{8} R_i}{8} = \frac{4.9-0.6-0.4}{8} = \frac{3.9}{8} = 0.4875 \text{ kilogram}$$

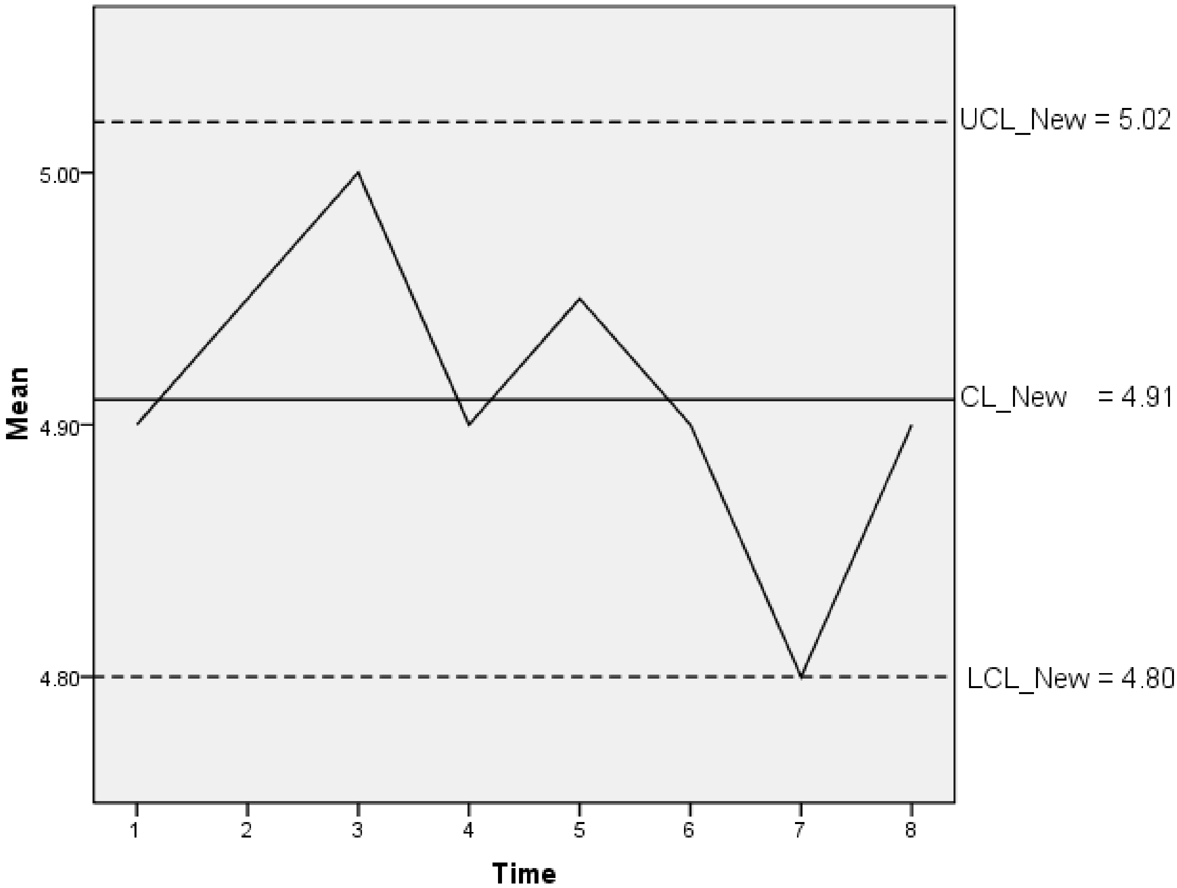
UCL$_{NEW}$ = $\bar{\bar{X}}_{NEW} + A_2\bar{R}_{NEW}$ = $4.91 + 0.223(0.4875)$ = 5.02 kilograms

CL$_{NEW}$ = $\bar{\bar{X}}_{NEW}$ = 4.91 kilograms

LCL$_{NEW}$ = $\bar{\bar{X}}_{NEW} - A_2\bar{R}_{NEW}$ = $4.91 - 0.223(0.4875)$ = 4.80 kilograms.

In this case $n = 15$, $A_2 = 0.223$

$\bar{X}$ **- Chart (Based on R)** Calculated without using points 6 and 8



From the control chart for the mean, it is found that after removing points 6 and 8 (original), the rice packing process is **in control.**

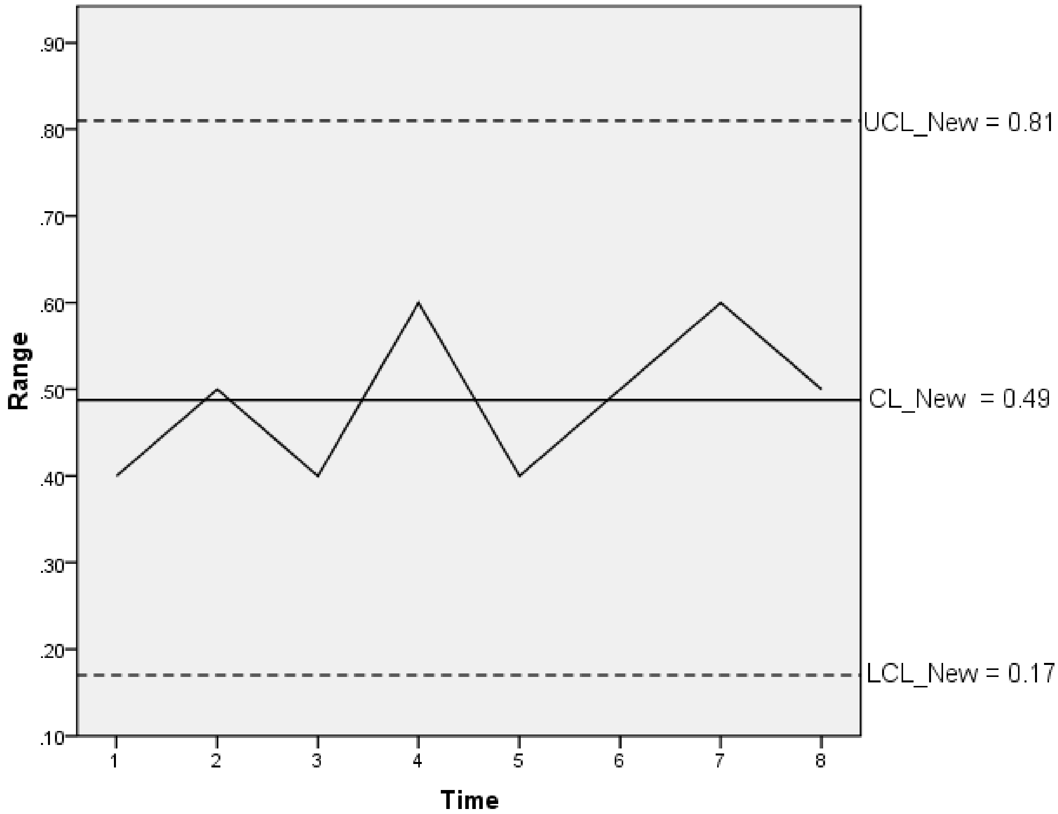
II. **R - Chart**: Calculated by discarding the data in production hours 6 and 8.

UCL$_{NEW}$ = $D_4\bar{R}_{NEW}$ = $1.652(0.4875)$ = 0.81 kilogram

$CL_{NEW} = \bar{R}_{NEW} = 0.4875 = 0.4875$ kilograms

$LCL_{NEW} = D_3\bar{R}_{NEW} = 0.348(0.4875) = 0.17$ kilogram

In this case $n=15$, $D_3 = 0.348$, $D_4 = 1.652$



From the range control chart, it is found that the distribution of the ranges of the weights of the packaged rice bags remains **in control.**

Conclusion after discarding the data in production hours 6 and 8
From Figure 1. $\bar{X}$ - Chart (Based on R) **(In Control)**
and Figure 2. R – Chart **(In Control)**

It can be concluded that the rice packing process using an automatic machine is **in control.**

Example 3.7 A bottle manufacturing factory produces 6–ounce glass bottles for sale in the market. The production control engineer randomly sampled bottles that passed through the production process over a period of 8 days, with an equal number each day, obtaining a total of 24 bottles, and recorded the average weight of the bottles (in oz.). The data obtained are as follows:

Mean weight of the bottles	5.7	6.0	5.5	6.0	5.9	6.4	5.8	5.9
Standard deviation	0.1	0.1	0.3	0.3	0.4	0.6	0.6	1.5

Given that **the sum** of the mean weights of the bottles = 47.2 oz.

and **the sum** of the standard deviations = 3.9 oz.

Construct the control chart for the means of weights of the 6 oz. bottles and the control chart for the dispersion of the weights of the 6 oz. bottles, and summarize the results obtained.

Example 3.8 In a tire production line, a researcher takes samples at 6 time intervals. At each time interval, 10 tires are randomly selected and the pressure (pounds per square inch) of each tire is measured. The data obtained are as follows:

Sample	1	2	3	4	5	6
Mean	32.4	35.1	34.6	32.4	33.4	35.1
R	3.8	2.1	3.7	2.2	3.0	3.1

Construct the appropriate control chart(s) for the data and summarize the results obtained.

3.2.2 Control Chart for Attributes

In inspecting product quality to determine whether each item meets the specified standard or not, if the proportion of nonconforming products is of interest, the control charts used are the P control chart (P–Chart) and the C control chart (C–Chart).

- **P Control Chart (Percent Defective Chart : P-Chart)** is a chart constructed to control the proportion of defective or damaged products (Defective Units) in a production lot, which consists of a large number of products.

Randomly select n_i units of products from the production process during a certain time period, with equal intervals between samplings of n_i periods (or an equal number of k subgroups), and then count the number of defective units in each of the n_i sampled units.

Time interval i	n	Number of Defective	$\hat{P}_i = \frac{x_i}{n_i}$
1 2 $\vdots$ k	n_1 n_2 $\vdots$ n_k	x_1 x_2 $\vdots$ x_k	$\hat{P}_1$ $\hat{P}_2$ $\vdots$ $\hat{P}_k$
Total	$\sum_{i=1}^{k} n_i$	$\sum_{i=1}^{k} x_i$	$\sum_{i=1}^{k} \hat{P}_i = \sum_{i=1}^{k} \frac{x_i}{n_i}$
$n_1 = n_2 = \ldots$	$= n_k = n$		

Consider the following case: $\boxed{n_1 = n_2 = \ldots = n_k} = n$

Let x_i denote the number of defectives in time interval i ; $i = 1, 2, \ldots, k$

n_i denote the sample size in time interval i

$\hat{P}_i$ denote the proportion of defective products in time interval i

Where $\hat{P}_i = \frac{x_i}{n_i} = \frac{x_i}{n}$; $i = 1, 2, \ldots, k$

Let x_i denote the number of defectives in time interval i from the n sampled units ; $x_i = 0, 1, 2, \ldots, n$

Thus, $x_i \sim Binomial\left(\mu = np, \sigma^2 = np(1-p)\right)$

where p = the proportion of defectives in the entire population ; $0 \le p \le 1$

and $\hat{P}_i$ = the proportion of defectives in **the sample** of n_i units in time interval i ; $0 \le \hat{P}_i \le 1$

where $\hat{P}_i = \frac{x_i}{n_i} = \frac{x_i}{n}$; $i = 1, 2, \ldots, k$

thus $\hat{P}_i \sim Binomial\left(\mu_{\hat{P}} = p, \sigma^2_{\hat{P}} = \frac{p(1-p)}{n}\right)$

When the production process is under control, the 3σ interval of $\hat{P}_i$ is $\mu_{\hat{P}} \pm 3\sigma_{\hat{P}}$

UCL = $\mu_{\hat{P}} + 3\sigma_{\hat{P}}$ = $p + 3\sqrt{\frac{p(1-p)}{n}}$

CL = $\mu_{\hat{P}}$ = p

LCL = $\mu_{\hat{P}} - 3\sigma_{\hat{P}}$ = $p - 3\sqrt{\frac{p(1-p)}{n}}$

In practice, the value of p (p is the proportion of defectives in the entire population) **is unknown** and is estimated by $\bar{P}$

where $\bar{P}$ denotes the mean of **the sample proportions over k time intervals**. There are 2 formulas used for computation. 1) $\bar{P} = \frac{\sum_{i=1}^{k} \hat{P}_i}{k} = \frac{\hat{P}_1 + \hat{P}_2 + \ldots + \hat{P}_k}{k}$ 2) $\bar{P} = \frac{\sum_{i=1}^{k} x_i}{nk} = \frac{x_1 + x_2 + \ldots + x_k}{nk}$

The 3σ interval of $\hat{P}_i$ is $\bar{P} \pm 3\sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$;

UCL = $\bar{P}+3\sqrt{\dfrac{\bar{P}\left(1-\bar{P}\right)}{n}}$
CL = $\bar{P}$
LCL = $\bar{P}-3\sqrt{\dfrac{\bar{P}\left(1-\bar{P}\right)}{n}}$

Note: If LCL < 0, then use LCL = 0 instead.

Example 3.9 A decorative light bulb factory produces a large number of bulbs according to orders from retail customers. Before sending the bulbs to the customers, the quality control department randomly selects 100 bulbs every production hour for 5 hours and tests whether they are defective or not. The data obtained are as follows:

Hour	Bulbs sampled	**Non-defective** bulbs	Defective bulbs
1	100	95	5
2	100	94	6
3	100	93	7
4	100	96	4
5	100	92	8
Total	500	470	30

Construct the control chart(s) to be used for inspecting the light bulb production process.

Solution Let x_i denote the number of defective light bulbs from the 100 bulbs sampled in each hour i ; $i=1,2,3,4,5$

In this case, the sample size of light bulbs randomly selected in each hour is 100 bulbs equally

$$\boxed{n_1 = n_2 = n_3 = n_4 = n_5} = 100 = n$$

That is, let $\hat{P}_i$ denote the proportion of defective light bulbs from the 100 bulbs sampled in hour i ; $i=1,2,3,4,5$

Compute the sample proportions of defective light bulbs from the 100 bulbs sampled in time intervals 1, 2, 3, 4, and 5 as follows

$$\hat{P}_1 = \frac{x_1}{n_1} = \frac{x_1}{n} = \frac{5}{100} = 0.05$$

$$\hat{P}_2 = \frac{x_2}{n_2} = \frac{x_2}{n} = \frac{6}{100} = 0.06$$

$$\hat{P}_3 = \frac{x_3}{n_3} = \frac{x_3}{n} = \frac{7}{100} = 0.07$$

$$\hat{P}_4 = \frac{x_4}{n_4} = \frac{x_4}{n} = \frac{4}{100} = 0.04$$

$$\hat{P}_5 = \frac{x_5}{n_5} = \frac{x_5}{n} = \frac{8}{100} = 0.08$$

Using the sample proportions of defective light bulbs in time intervals 1, 2, 3, 4, and 5, compute **the mean of the sample proportions of the defective light bulbs sampled over the 5 hours as follows**

1) $$\bar{P} = \frac{\sum_{i=1}^{k}\hat{P}_i}{k} = \frac{\sum_{i=1}^{5}\hat{P}_i}{5} = \frac{\hat{P}_1+\hat{P}_2+\hat{P}_3+\hat{P}_4+\hat{P}_5}{5}$$

$$= \frac{0.05+0.06+0.07+0.04+0.08}{5} = \frac{0.30}{5} = 0.06$$

Alternatively, the mean of the sample proportions of **the defective light bulbs sampled over the 5 hours** can be computed from the following formula

2) $$\bar{P} = \frac{\sum_{i=1}^{k}x_i}{nk} = \frac{\sum_{i=1}^{5}x_i}{100\times 5} = \frac{x_1+x_2+x_3+x_4+x_5}{100\times 5}$$

$$= \frac{5+6+7+4+8}{100\times 5} = \frac{30}{500} = 0.06$$

In sampling 100 light bulbs per hour for 5 consecutive hours, it is found that the average proportion of defective bulbs is 0.06, or the average number of defective bulbs is 6 percent of the bulbs sampled over the 5 hours

Constructing the P-Chart for the proportion of defective light bulbs uses the following formulae to calculate the Upper Control Limit (UCL) and Lower Control Limit (LCL)

$$\text{UCL} = \bar{P} + 3\sqrt{\frac{\bar{P}\left(1-\bar{P}\right)}{n}} = 0.06 + 3\sqrt{\frac{0.06\left(1-0.06\right)}{100}}$$

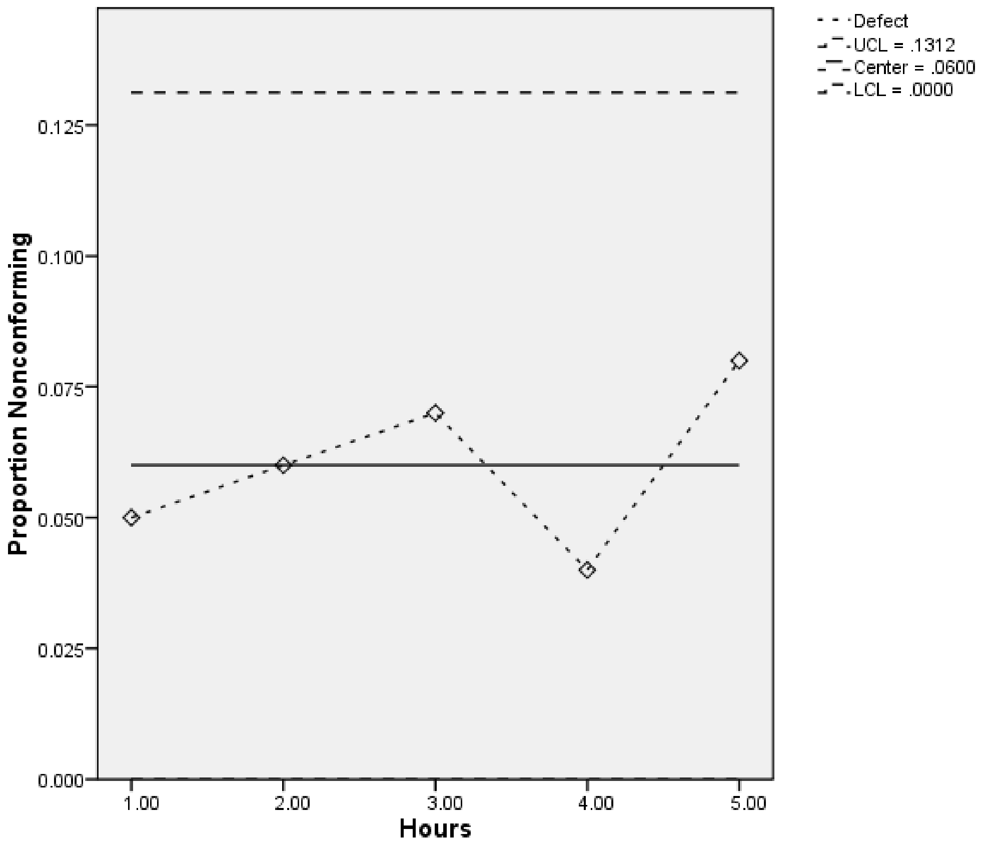
$$= 0.06 + 0.0712 = 0.1312$$

$$\text{CL} = \bar{P} = 0.06$$

$$\text{LCL} = \bar{P} - 3\sqrt{\frac{\bar{P}\left(1-\bar{P}\right)}{n}} = 0.06 - 3\sqrt{\frac{0.06\left(1-0.06\right)}{100}}$$

$$= 0.06 - 0.0712 = -0.0112$$

Since the value LCL = -0.0112 < 0, the LCL is adjusted to 0 and this value is used instead



From the P–Chart obtained, it is found that no points lie outside the UCL and LCL.

Therefore, it can be concluded that the decorative light bulb production process is **in control.**

Example 3.10 A factory producing 6–ounce canned fruit juice wants to determine whether the machine is malfunctioning. The production control department randomly sampled 50 cans every half hour of production and counted the number of defective cans. The data obtained are as follows

Sample set number	Number of Defective cans	Proportion of defective cans	Sample set number	Number of defective cans	Proportion of defective cans
1	12	0.24	16	8	0.16
2	15	0.30	17	10	0.20
3	8	0.16	18	5	0.10
4	10	0.20	19	13	0.26
5	4	0.08	20	11	0.22
6	7	0.14	21	20	0.40
7	16	0.32	22	18	0.36
8	9	0.18	23	24	0.48
9	14	0.28	24	15	0.30
10	10	0.20	25	9	0.18
11	5	0.10	26	12	0.24
12	6	0.12	27	7	0.14
13	17	0.34	28	13	0.26
14	12	0.24	29	9	0.18
15	22	0.44	30	6	0.12

Construct the control chart to determine whether the canned fruit juice filling process is under control.

Solution Let x_i denote the number of defective cans from the 50 sampled cans of fruit juice in time interval i ; $i = 1, 2, \ldots, 30$

In this case, the sample size of the canned fruit juice selected in each time interval is 50 cans equally

$$\boxed{n_1 = n_2 = \ldots = n_{30}} = 50 = n$$

That is, let $\hat{P}_i$ denote the proportion of defective cans from the 50 sampled cans of fruit juice in time interval i ; $i = 1, 2, \ldots, 30$

Compute the sample proportions of defective cans from the 50 sampled cans in time intervals 1, 2, … , 30 as follows

$\hat{P}_1 = \frac{x_1}{n} = \frac{12}{50} = 0.24$; $\hat{P}_{16} = \frac{x_{16}}{n} = \frac{8}{50} = 0.16$

$\hat{P}_2 = \frac{x_2}{n} = \frac{15}{50} = 0.30$; $\hat{P}_{17} = \frac{x_{17}}{n} = \frac{10}{50} = 0.20$

$\hat{P}_3 = \frac{x_3}{n} = \frac{8}{50} = 0.16$; $\hat{P}_{18} = \frac{x_{18}}{n} = \frac{5}{50} = 0.10$

$\hat{P}_4 = \frac{x_4}{n} = \frac{10}{50} = 0.20$; $\hat{P}_{19} = \frac{x_{19}}{n} = \frac{13}{50} = 0.26$

$\hat{P}_5 = \frac{x_5}{n} = \frac{4}{50} = 0.08$; $\hat{P}_{20} = \frac{x_{20}}{n} = \frac{11}{50} = 0.22$

$\hat{P}_6 = \frac{x_6}{n} = \frac{7}{50} = 0.14$; $\hat{P}_{21} = \frac{x_{21}}{n} = \frac{20}{50} = 0.40$

$\hat{P}_7 = \frac{x_7}{n} = \frac{16}{50} = 0.32$; $\hat{P}_{22} = \frac{x_{22}}{n} = \frac{18}{50} = 0.36$

$\hat{P}_8 = \frac{x_8}{n} = \frac{9}{50} = 0.18$; $\hat{P}_{23} = \frac{x_{23}}{n} = \frac{24}{50} = 0.48$

$\hat{P}_9 = \frac{x_9}{n} = \frac{14}{50} = 0.28$; $\hat{P}_{24} = \frac{x_{24}}{n} = \frac{15}{50} = 0.30$

$\hat{P}_{10} = \frac{x_{10}}{n} = \frac{10}{50} = 0.20$; $\hat{P}_{25} = \frac{x_{25}}{n} = \frac{9}{50} = 0.18$

$\hat{P}_{11} = \frac{x_{11}}{n} = \frac{5}{50} = 0.10$; $\hat{P}_{26} = \frac{x_{26}}{n} = \frac{12}{50} = 0.24$

$\hat{P}_{12} = \frac{x_{12}}{n} = \frac{6}{50} = 0.12$; $\hat{P}_{27} = \frac{x_{27}}{n} = \frac{7}{50} = 0.14$

$\hat{P}_{13} = \frac{x_{13}}{n} = \frac{17}{50} = 0.34$; $\hat{P}_{28} = \frac{x_{28}}{n} = \frac{13}{50} = 0.26$

$\hat{P}_{14} = \frac{x_{14}}{n} = \frac{12}{50} = 0.24$; $\hat{P}_{29} = \frac{x_{29}}{n} = \frac{9}{50} = 0.18$

$\hat{P}_{15} = \frac{x_{15}}{n} = \frac{22}{50} = 0.44$; $\hat{P}_{30} = \frac{x_{30}}{n} = \frac{6}{50} = 0.12$

Using the sample proportions of defective cans in time intervals 1, 2, ..., 30, compute

<u>The mean of the sample proportions of defective cans sampled over the 30 time</u> intervals as follows

1) $$\bar{P} = \frac{\sum_{i=1}^{k}\hat{P}_i}{k} = \frac{\sum_{i=1}^{30}\hat{P}_i}{30} = \frac{\hat{P}_1+\hat{P}_2+\ldots+\hat{P}_{30}}{30}$$

$$= \frac{0.24+0.30+\ldots+0.12}{30} = \frac{6.94}{30} = 0.2313$$

Alternatively, **the mean of the sample proportions of defective cans sampled over the 30 time** intervals can be calculated using the following formula

2) $$\bar{P} = \frac{\sum_{i=1}^{k}x_i}{nk} = \frac{\sum_{i=1}^{30}x_i}{50\times 30} = \frac{x_1+x_2+\ldots+x_{30}}{50\times 30}$$

$$= \frac{12+15+\ldots+6}{50\times 30} = \frac{347}{1500} = 0.2313$$

From sampling 50 cans of fruit juice every half hour for 30 consecutive time intervals, it is found that the average proportion of defective canned fruit juice is 0.2313, or the average number of defective fruit juice cans is 23.13 percent of the 50 cans sampled over the 30 intervals.

Constructing the P-Chart for the proportion of defective canned fruit juice uses the following formulae for calculating the Upper Control Limit (UCL) and Lower Control Limit (LCL)

$$\text{UCL} = \bar{P}+3\sqrt{\frac{\bar{P}\left(1-\bar{P}\right)}{n}} = 0.2313+3\sqrt{\frac{0.2313(1-0.2313)}{50}}$$

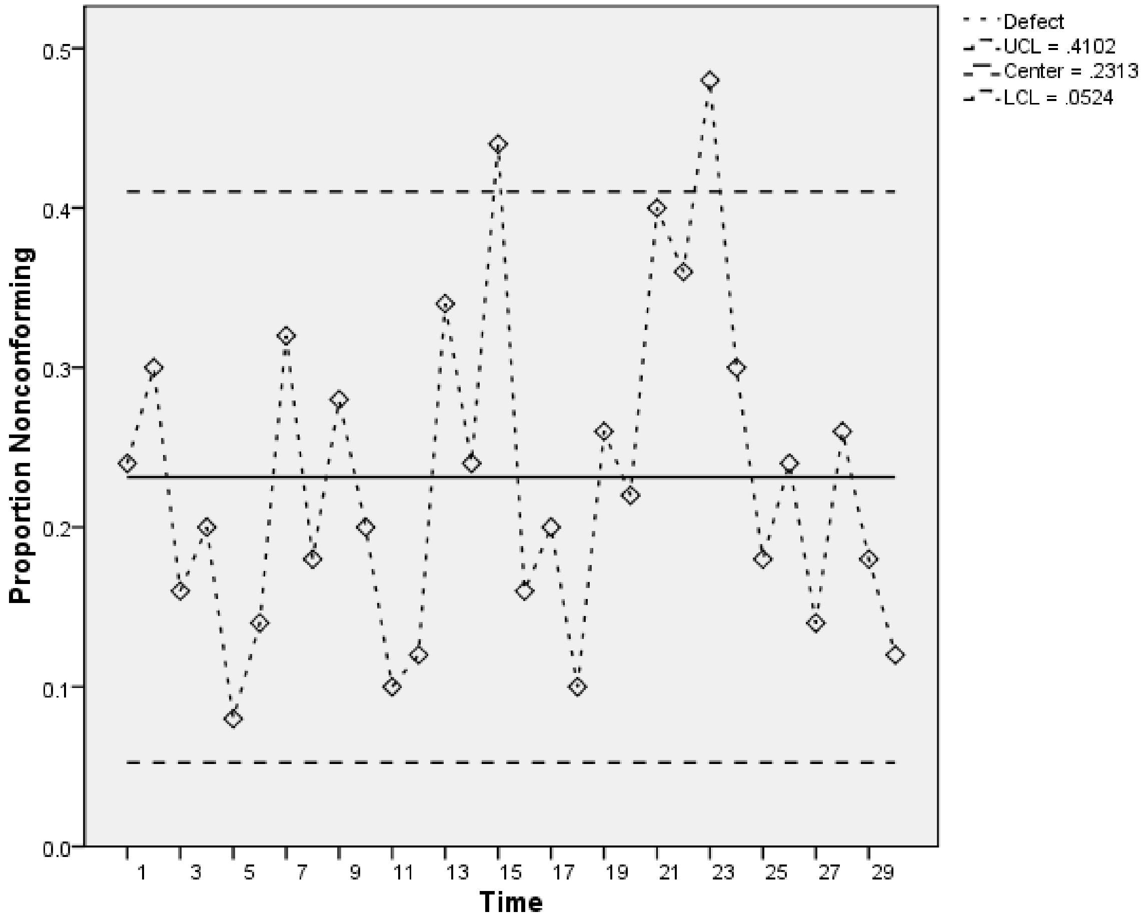
$$= 0.2313 + 0.1789 = 0.4102$$

$$\text{CL} = \bar{P} = 0.2313$$

$$\text{LCL} = \bar{P}-3\sqrt{\frac{\bar{P}\left(1-\bar{P}\right)}{n}} = 0.2313-3\sqrt{\frac{0.2313(1-0.2313)}{50}}$$

$$= 0.2313 - 0.1789 = 0.0524$$

Since LCL = -0.0112 < 0, the LCL is adjusted to 0 and this value is used instead



From the P–Chart obtained, it is found that at time interval 15 and time interval 23, there are points outside the UCL.

Therefore, it can be concluded that the canned fruit juice production process is **out of control**, and production should be stopped in order to inspect the machine used for filling the cans of fruit juice.

Example 3.11 A factory producing fuses used in computers wants to check whether its production is of good quality. The quality control department randomly samples the fuses produced for inspection every hour in equal numbers for 12 consecutive hours, totaling 600 pieces. After inspection, it is found that the numbers of unusable fuses in each hour are 3, 2, 5, 1, 0, 3, 2, 5, 3, 2, 8, and 10 pieces, respectively. Construct the appropriate control chart(s) to be used for inspecting the fuse production of this factory and summarize the results obtained.

- **<u>C Control Chart</u> (Chart for Number of Defects per Sampling Unit : C - Chart)** is a chart constructed to control the number of defects per sampling unit of a product. A sample of k product units is taken, and in each sampling unit, the number of defects is k.

Sampling unit no.	Number of defects per sampling unit
1 2 $\vdots$ k	c_1 c_2 $\vdots$ c_k
Total	$\sum_{i=1}^{k} c_i$

Let c_i denote **the number of defects per sampling unit of product <u>i</u>** ; $c_i = 0,1,2,\ldots$, thus $c_i \sim Poisson\left(\mu_c, \sigma_c^2 = \mu_c\right)$

When the production process is under control, the 3σ interval of c_i is $\mu_c \pm 3\sigma_c$.

UCL = $\mu_c + 3\sigma_c$ = $\mu_c + 3\sqrt{\mu_c}$

CL = μ_c

LCL = $\mu_c - 3\sigma_c$ = $\mu_c - 3\sqrt{\mu_c}$

Where μ_c **<u>denotes the population mean of the number of defects per 1 product unit</u>**

In practice, the value of μ_c **<u>is unknown</u>** and is estimated by $\bar{C}$

Let $\bar{C}$ **<u>denote the average number of defects per 1 product unit</u>**

$$\bar{C} = \frac{\sum_{i=1}^{k} c_i}{k} \quad ;$$

Thus, the 3σ interval of c_i is $\bar{C} \pm 3\sqrt{\bar{C}}$;

$$\begin{aligned} UCL &= \bar{C}+3\sqrt{\bar{C}} \\ CL &= \bar{C} \\ LCL &= \bar{C}-3\sqrt{\bar{C}} \end{aligned}$$

Note: If LCL < 0, then LCL = 0 is used instead

Example 3.12 A factory producing self-assembly clothes drying racks, packaged with separated parts and assembly instructions for customers, received complaints stating that some parts were defective or missing, making it impossible to assemble. The manufacturer therefore sampled 10 boxes of the produced products and found the following numbers of defective parts:

Box no.	Number of defects, Locations
1	8
2	7
3	8
4	4
5	6
6	4
7	3
8	6
9	5
10	9

Construct the C - Chart

Solution Let c_i denote the number of defective parts of the clothes drying rack in box i ; $i=1,2,\ldots,10$. In this case, 10 boxes of clothes drying racks were sampled, and the number of defective parts in each box was counted as follows

$c_1 = 8$ locations ; $c_6 = 4$ locations

$c_2 = 7$ locations ; $c_7 = 3$ locations

$c_3 = 8$ locations ; $c_8 = 6$ locations

$c_4 = 4$ locations ; $c_9 = 5$ locations

$c_5 = 6$ locations ; $c_{10} = 9$ locations

Using the number of defective parts in each box, **compute the mean number of defective parts per box** as follows

$$\bar{C} = \frac{\sum_{i=1}^{k} c_i}{k} = \frac{\sum_{i=1}^{10} c_i}{10} = \frac{c_1 + c_2 + \ldots + c_{10}}{10} = \frac{8 + 7 + \ldots + 9}{10} = \frac{60}{10} = 6 \text{ locations}$$

On average, the sampled boxes of clothes drying racks contain 6 defective parts per box

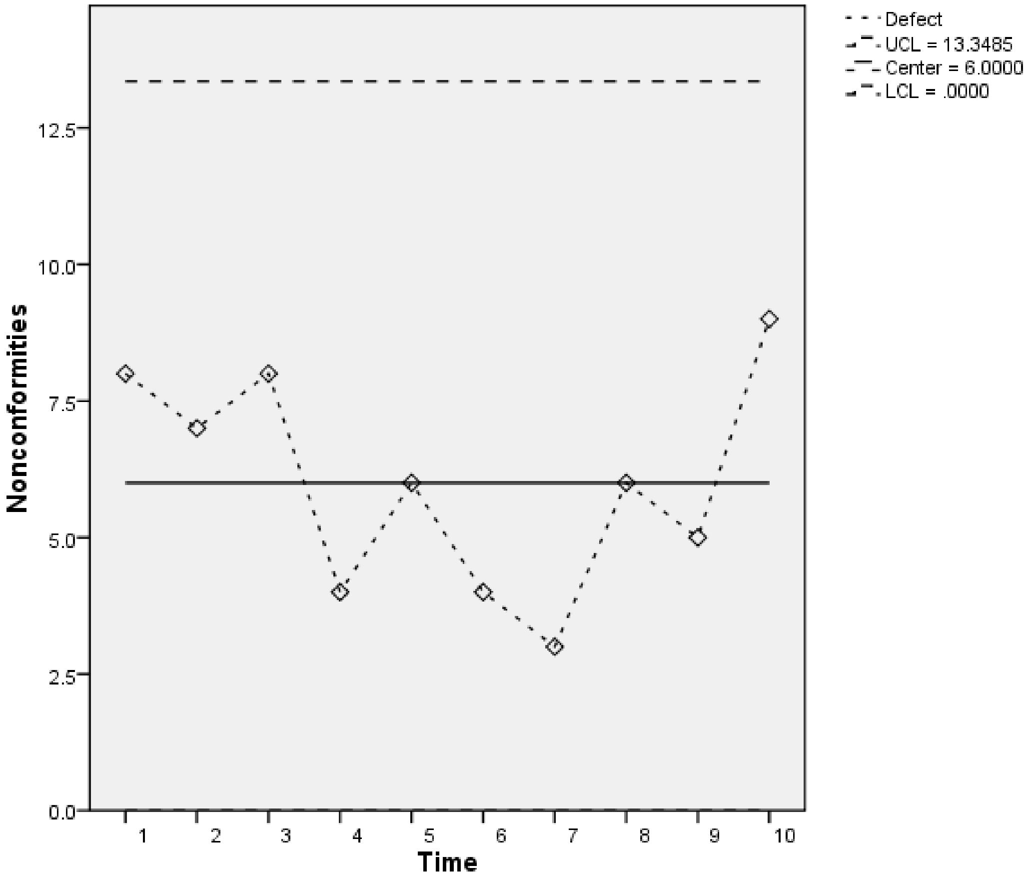
Constructing the control chart for the number of defective parts in the boxed clothes drying racks (C–Chart) uses the following formulae for the Upper Control Limit (UCL) and Lower Control Limit (LCL)

UCL = $\bar{C} + 3\sqrt{\bar{C}}$ = $6 + 3\sqrt{6}$ = 6 + 7.3485 = 13.3485 locations

CL = $\bar{C}$ = 6 locations

LCL = $\bar{C} - 3\sqrt{\bar{C}}$ = $6 - 3\sqrt{6}$ = 6 – 7.3485 = -1.3485 locations

Since LCL = -1.3485 < 0, the LCL is adjusted to 0 and this value is used instead



From the resulting C–Chart, it is found that none of the points lie outside the UCL and LCL

Therefore, it can be concluded that the boxed clothes drying rack production process is **in control.**

Example 3.13 A computer equipment manufacturing company produces mainboard components. The technical department randomly sampled 26 mainboards, and each one was inspected for defects. The data obtained are as follows

Unit no.	Number of defect locations	Unit no.	Number of defect locations
1	21	14	19
2	24	15	10
3	16	16	17
4	12	17	13
5	15	18	22

6	5	19	18
7	28	20	39
8	20	21	30
9	31	22	24
10	25	23	16
11	20	24	19
12	24	25	17
13	16	26	15

Construct the control chart to determine whether the mainboard production process is under control.

Solution Let c_i denote the number of defect locations on mainboard no. i ; $i = 1, 2, \ldots, 26$

In this case, 26 mainboards were sampled, and the number of defect locations in each piece was counted as follows

$c_1 = 21$ locations ;	$c_{11} = 20$ locations ;	$c_{21} = 30$ locations
$c_2 = 24$ locations ;	$c_{12} = 24$ locations ;	$c_{22} = 24$ locations
$c_3 = 16$ locations ;	$c_{13} = 16$ locations ;	$c_{23} = 16$ locations
$c_4 = 12$ locations ;	$c_{14} = 19$ locations ;	$c_{24} = 19$ locations
$c_5 = 15$ locations ;	$c_{15} = 10$ locations ;	$c_{25} = 17$ locations
$c_6 = 5$ locations ;	$c_{16} = 17$ locations ;	$c_{26} = 15$ locations
$c_7 = 28$ locations ;	$c_{17} = 13$ locations	
$c_8 = 20$ locations ;	$c_{18} = 22$ locations	
$c_9 = 31$ locations ;	$c_{19} = 18$ locations	
$c_{10} = 25$ locations ;	$c_{20} = 39$ locations	

Using the number of defect locations of each mainboard, **compute the mean number of defect locations per piece** as follows

$$\bar{C} = \frac{\sum_{i=1}^{k} c_i}{k} = \frac{\sum_{i=1}^{26} c_i}{26} = \frac{c_1 + c_2 + \ldots + c_{26}}{26} = \frac{21 + 24 + \ldots + 15}{26} = \frac{516}{26} = 19.8462$$

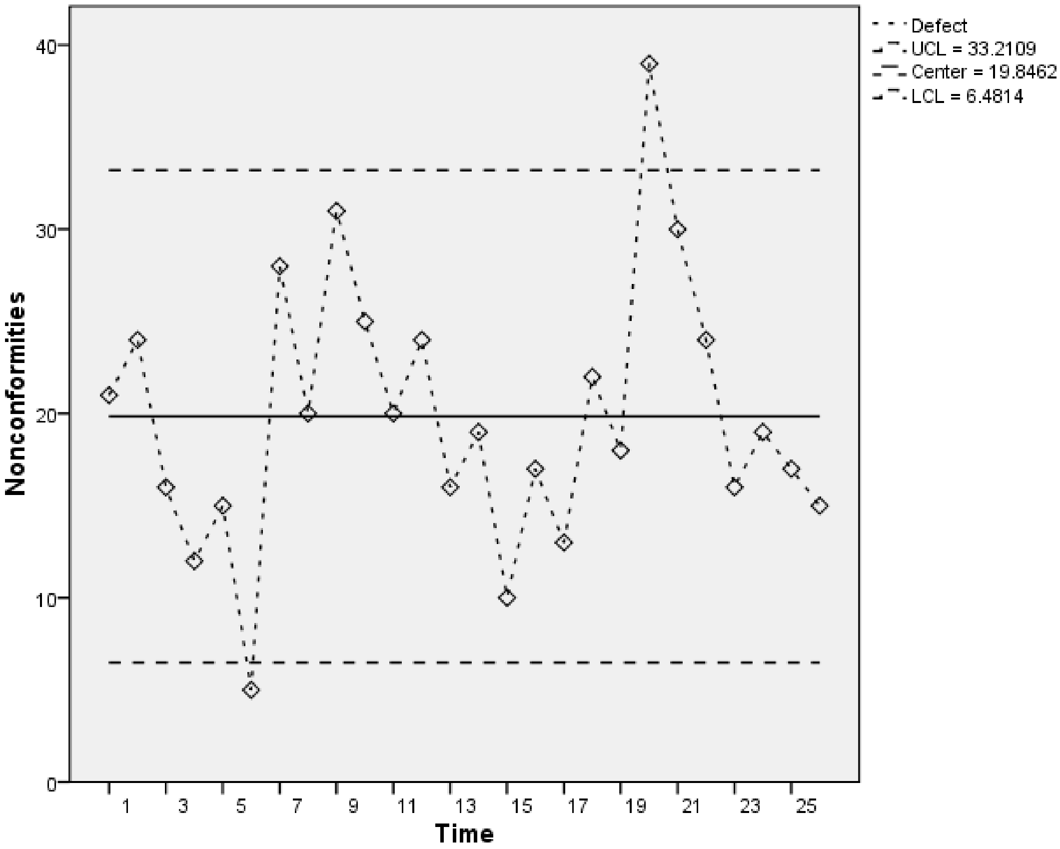
On average, the sampled mainboards have 19.85 defect locations per piece.

Constructing the control chart for the number of defect locations on mainboards (C–Chart) uses the following formulae for the Upper Control Limit (UCL) and Lower Control Limit (LCL)

UCL = $\bar{C} + 3\sqrt{\bar{C}}$ = $19.8462 + 3\sqrt{19.8462}$ = 33.2109 locations

CL = $\bar{C}$ = 19.8462 locations

LCL = $\bar{C} - 3\sqrt{\bar{C}}$ = $19.8462 - 3\sqrt{19.8462}$ = 6.4814 locations



From the C–Chart obtained, time intervals 6 and 20 have points lying outside the LCL and UCL, respectively. Therefore, it can be concluded that the mainboard production process is **out of control**, and production should be stopped to inspect the machine used in production.

Example 3.14 A fertilizer pellet factory wants to inspect the quality of the pellets that pass through the production process using the factory's machinery. The department head planned data collection by randomly selecting fertilizer from 1 container (area of 1 square meter) in each production hour for a total of 10 hours, and then counting the number of defects. The data obtained are as follows:

Hour	1	2	3	4	5	6	7	8	9	10
Number of defects (locations)	14	3	9	10	2	12	3	4	11	X

If the average number of defects over the 10 hours is 7.6, determine the number of defects in the hour with the highest defect count within the 10-hour period for use in the appropriate control chart for future monitoring of the production process, and summarize the result obtained.

Summary Table of Control Chart Formulae

Control Chart	LCL	CL	UCL
1. $\bar{X}$ – Chart (Based on R) R – Chart	$\bar{\bar{X}} - A_2\bar{R}$ $D_3\bar{R}$	$\bar{\bar{X}}$ $\bar{R}$	$\bar{\bar{X}} + A_2\bar{R}$ $D_4\bar{R}$
2. $\bar{X}$ – Chart (Based on S) S – Chart	$\bar{\bar{X}} - A_3\bar{S}$ $B_3\bar{S}$	$\bar{\bar{X}}$ $\bar{S}$	$\bar{\bar{X}} + A_3\bar{S}$ $B_4\bar{S}$
3. P – Chart	$\bar{P} - 3\sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$	$\bar{P}$	$\bar{P} + 3\sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$
4. C – Chart	$\bar{C} - 3\sqrt{\bar{C}}$	$\bar{C}$	$\bar{C} + 3\sqrt{\bar{C}}$

Table of Control Chart Constants

Sample Size = m	X-bar Chart Constants A_2	A_3	for sigma estimate d_2	R Chart Constants D_3	D_4	S Chart Constants B_3	B_4
2	1.880	2.659	1.128	0	3.267	0	3.267
3	1.023	1.954	1.693	0	2.574	0	2.568
4	0.729	1.628	2.059	0	2.282	0	2.266
5	0.577	1.427	2.326	0	2.114	0	2.089
6	0.483	1.287	2.534	0	2.004	0.030	1.970
7	0.419	1.182	2.704	0.076	1.924	0.118	1.882
8	0.373	1.099	2.847	0.136	1.864	0.185	1.815
9	0.337	1.032	2.970	0.184	1.816	0.239	1.761
10	0.308	0.975	3.078	0.223	1.777	0.284	1.716
11	0.285	0.927	3.173	0.256	1.744	0.321	1.679
12	0.266	0.886	3.258	0.283	1.717	0.354	1.646
13	0.249	0.850	3.336	0.307	1.693	0.382	1.618
14	0.235	0.817	3.407	0.328	1.672	0.406	1.594
15	0.223	0.789	3.472	0.347	1.653	0.428	1.572
16	0.212	0.763	3.532	0.363	1.637	0.448	1.552
17	0.203	0.739	3.588	0.378	1.622	0.466	1.534
18	0.194	0.718	3.640	0.391	1.608	0.482	1.518
19	0.187	0.698	3.689	0.403	1.597	0.497	1.503
20	0.180	0.680	3.735	0.415	1.585	0.510	1.490
21	0.173	0.663	3.778	0.425	1.575	0.523	1.477
22	0.167	0.647	3.819	0.434	1.566	0.534	1.466
23	0.162	0.633	3.858	0.443	1.557	0.545	1.455
24	0.157	0.619	3.895	0.451	1.548	0.555	1.445
25	0.153	0.606	3.931	0.459	1.541	0.565	1.435

Source: http://www.bessegato.com.br/UFJF/resources/table_of_control_chart_constants_old.pdf